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Kahan et al.

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(54) **VARIABLE INLET GUIDE VANE
APPARATUS AND COMPRESSOR
INCLUDING SAME**

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F04D 17/10 (2006.01)

F04D 29/056 (2006.01)

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(2013.01); **F04D 29/056** (2013.01); **F04D**
29/063 (2013.01); **F04D 29/4206** (2013.01);
F04D 29/624 (2013.01)

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F04D 29/4206; F04D 29/642; F04D
17/10; F04D 17/162; F04D 29/4213;
F05D 2250/51; F05D 2260/53; F05D
2260/523

See application file for complete search history.

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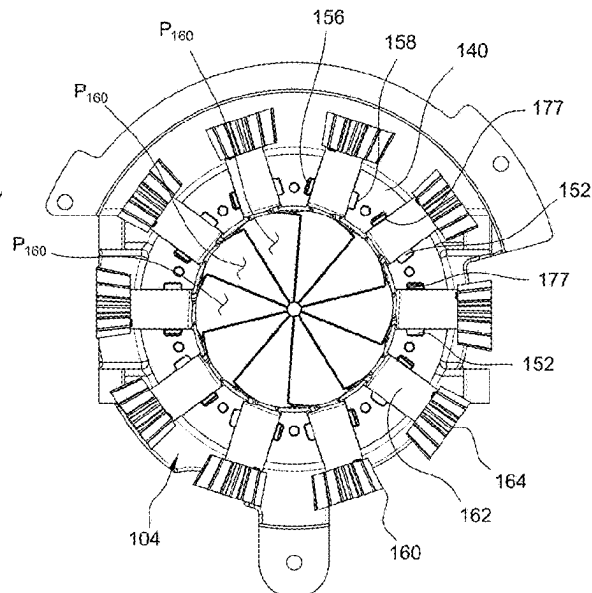
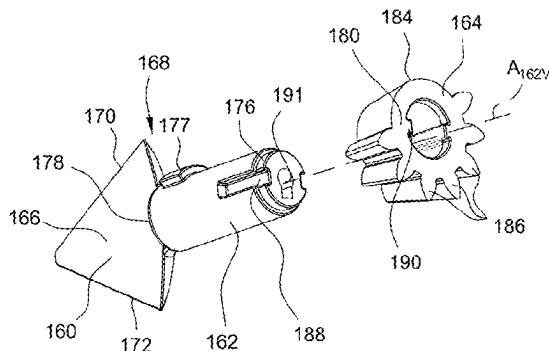
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(57)

ABSTRACT

A variable inlet guide vane apparatus for imparting a motion to a fluid flow entering into a compressor. The variable inlet guide vane apparatus includes a housing defining a fluid flow passageway. The housing includes a first and second housing portion, a ring gear, and a plurality of guide vanes. The guide vanes include a stem, a gear, and a vane. The stem is disposed between the first and second housing portions, the vane is arranged within the fluid flow passageway, and the gears are arranged exterior the housing.

20 Claims, 19 Drawing Sheets



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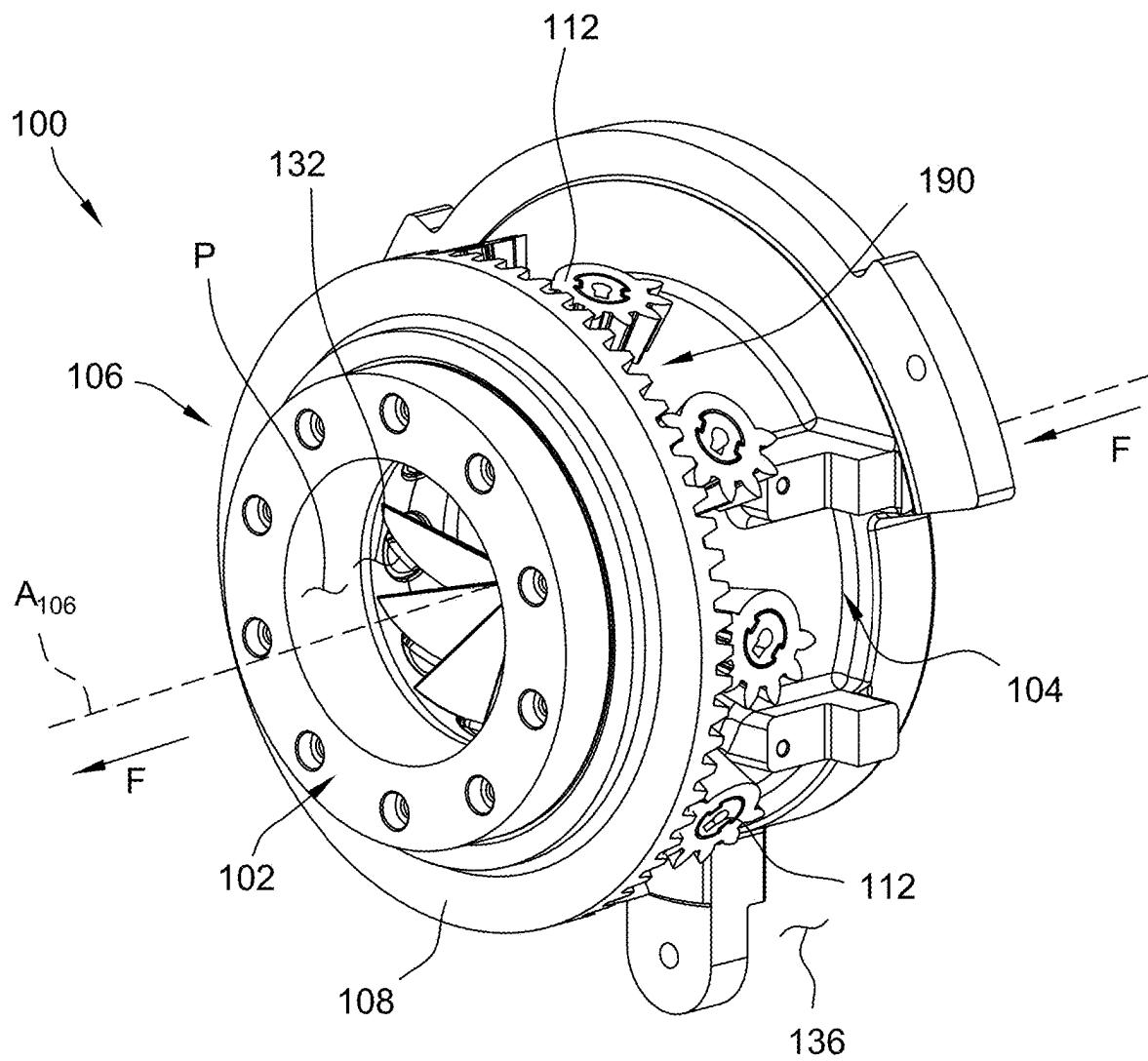


FIG. 1

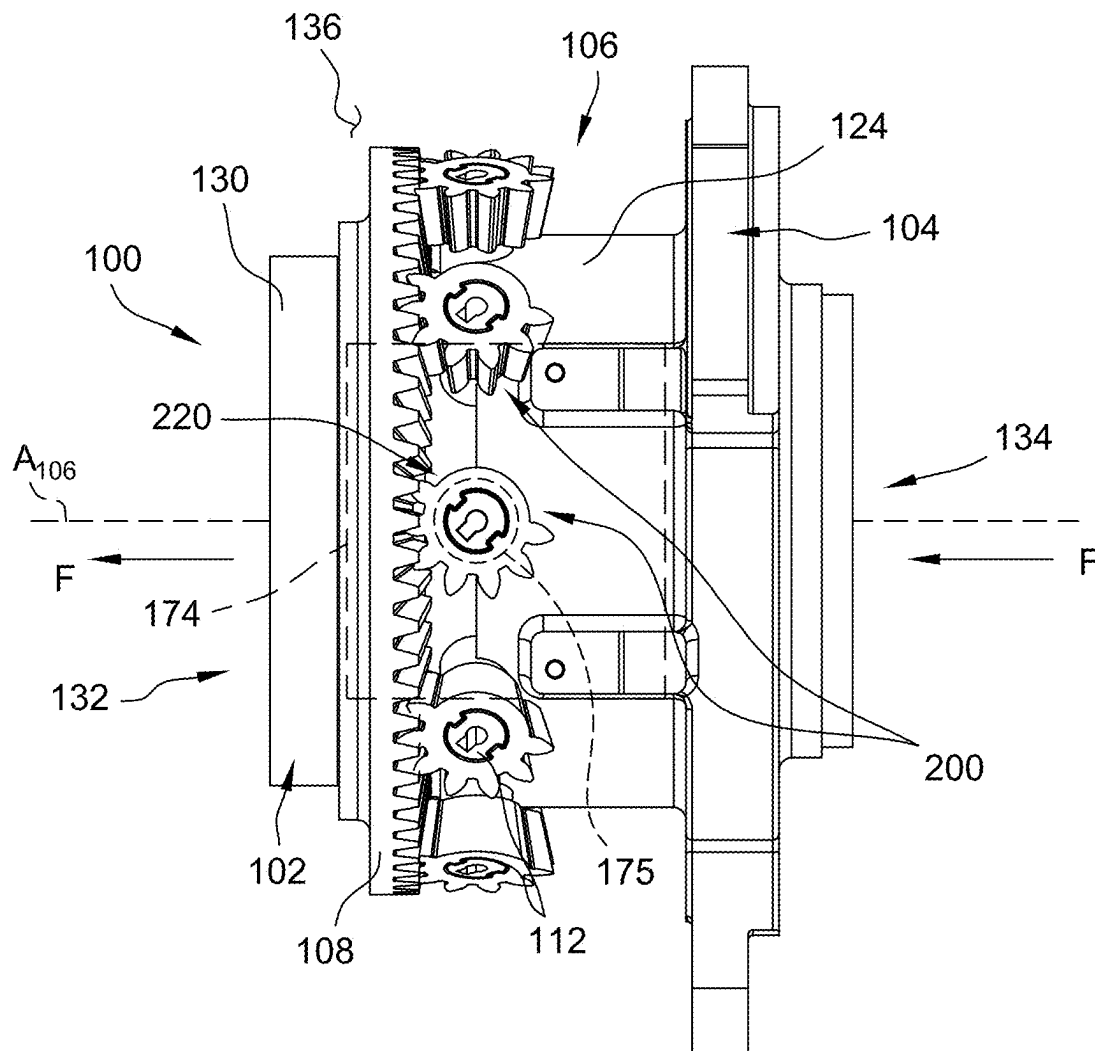


FIG. 2

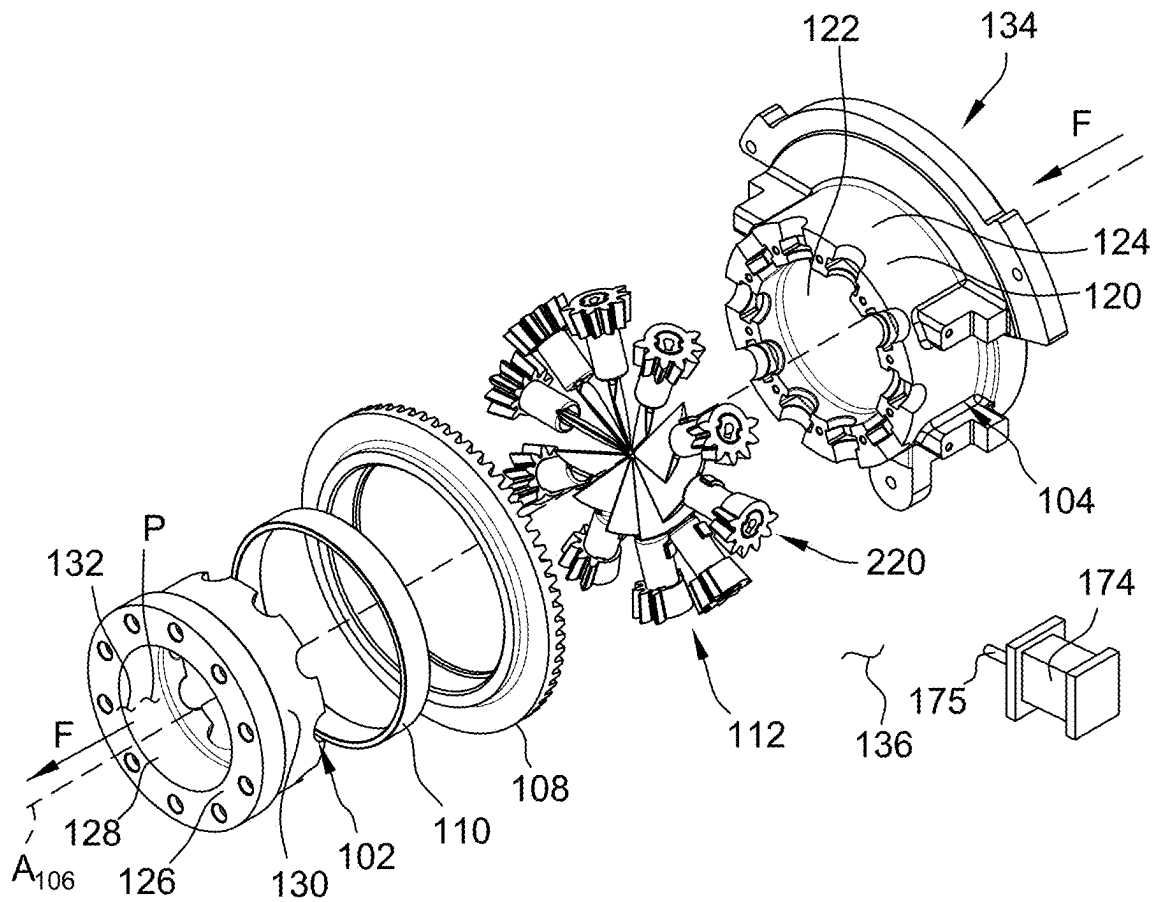


FIG. 3

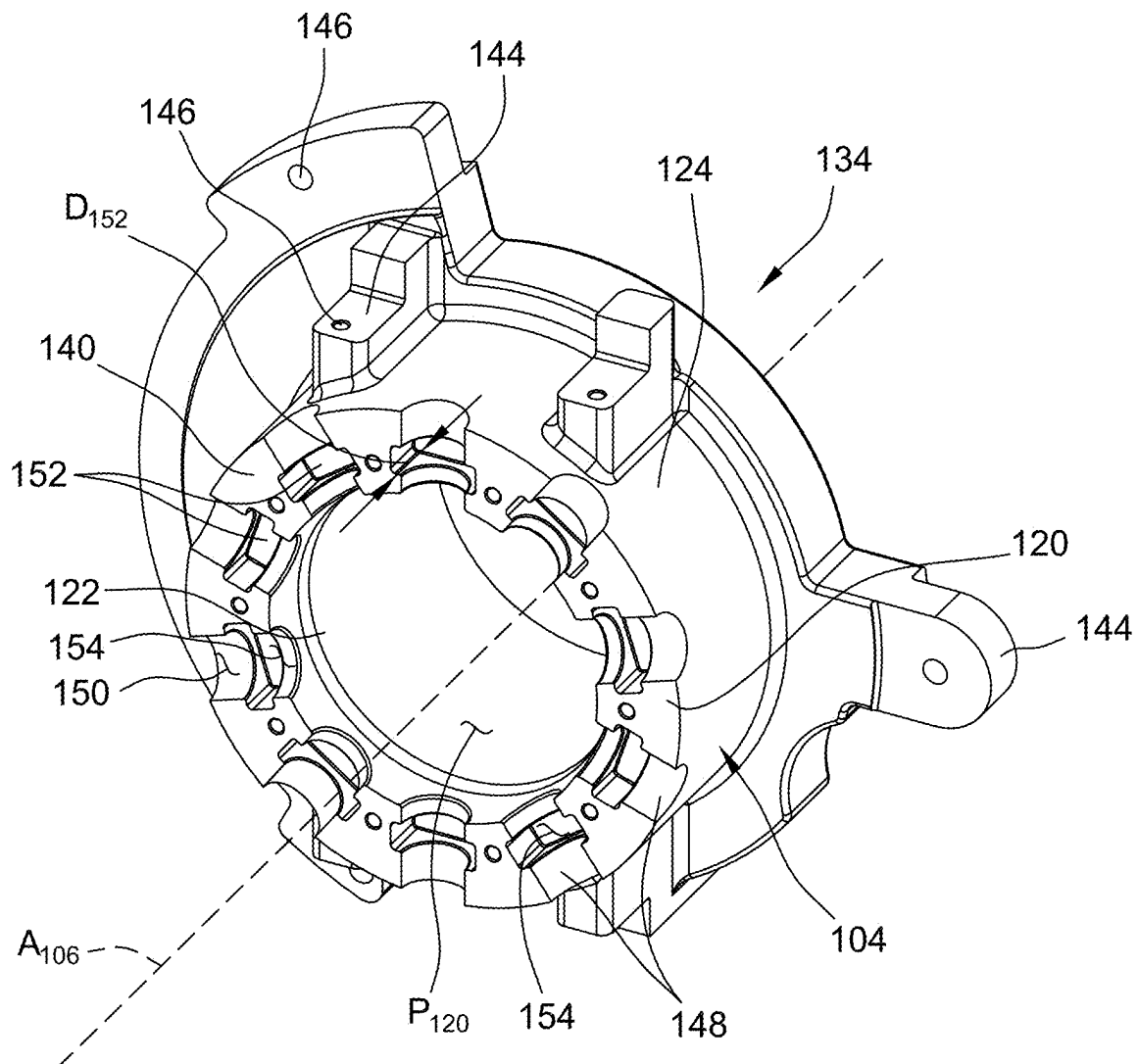


FIG. 4

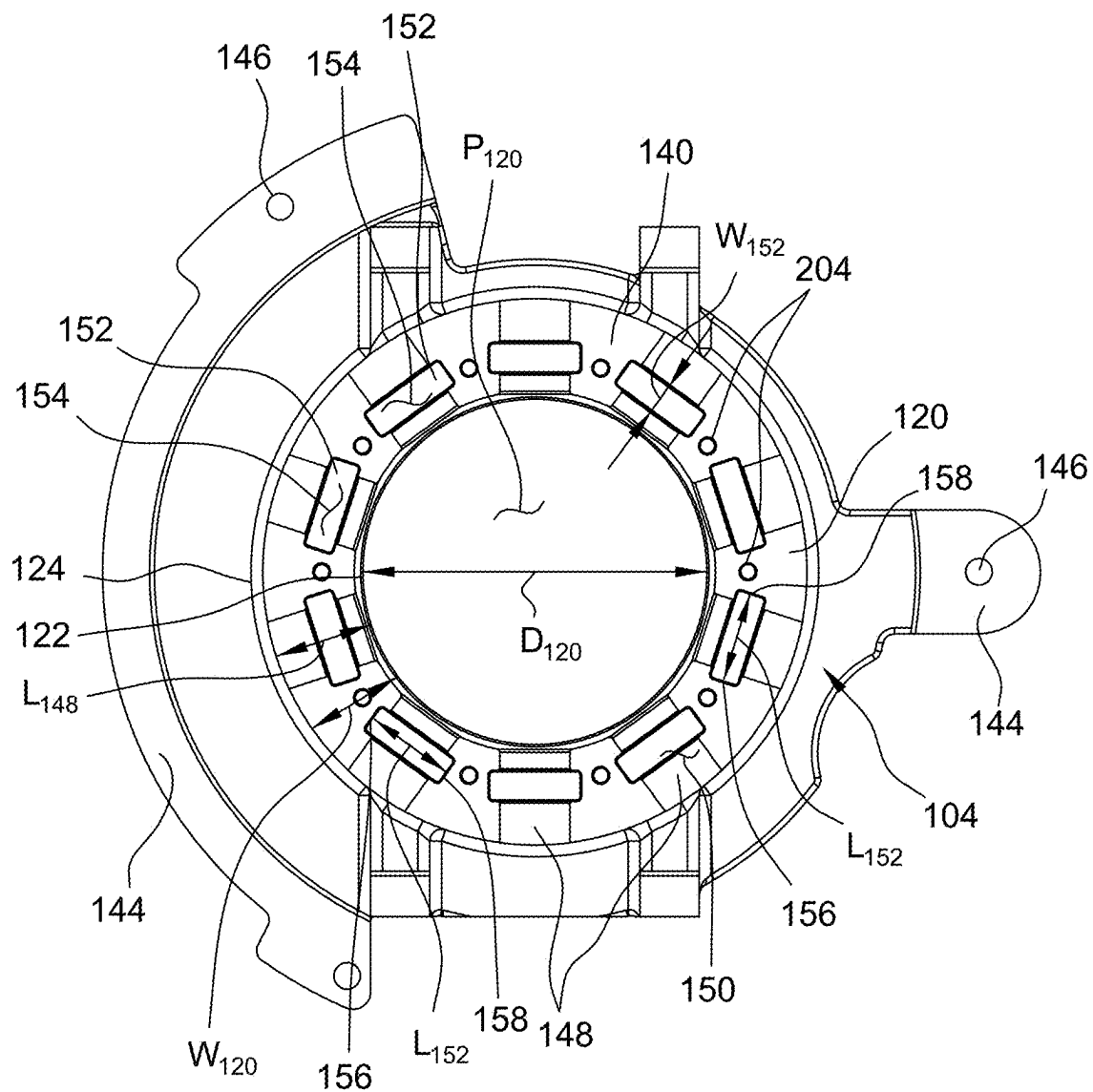


FIG. 5

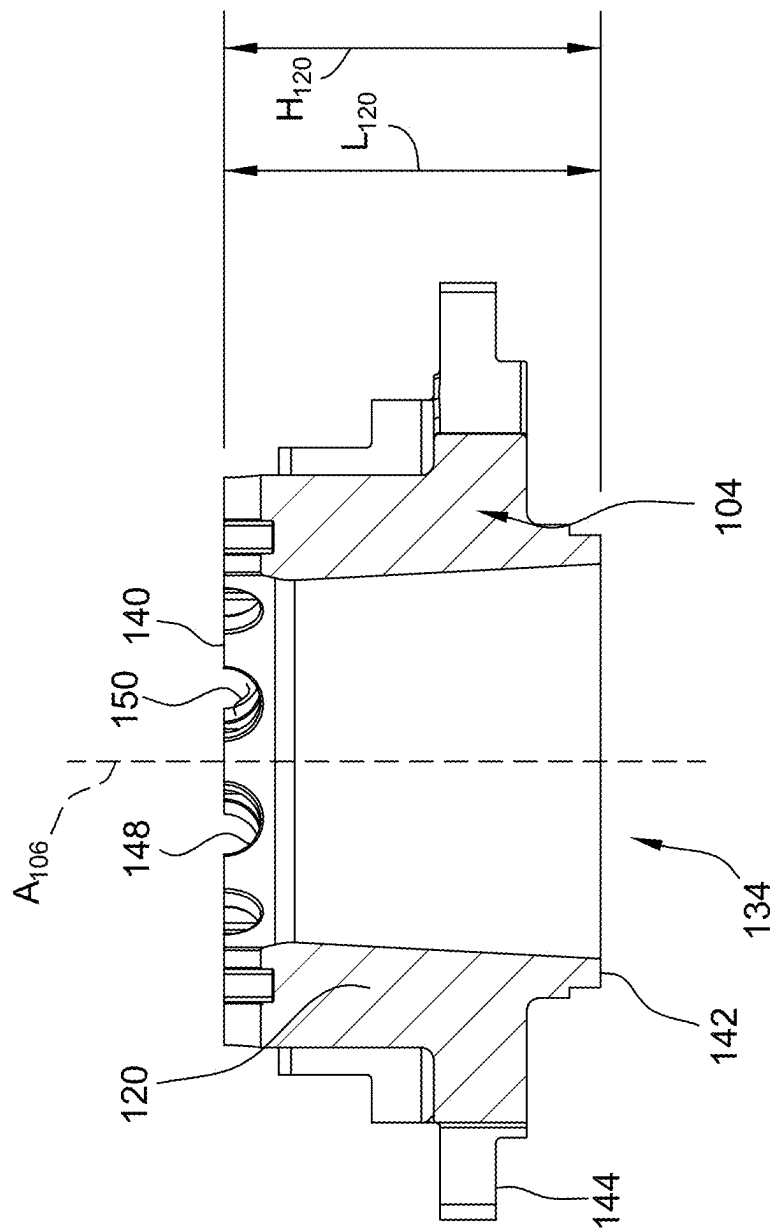


FIG. 6

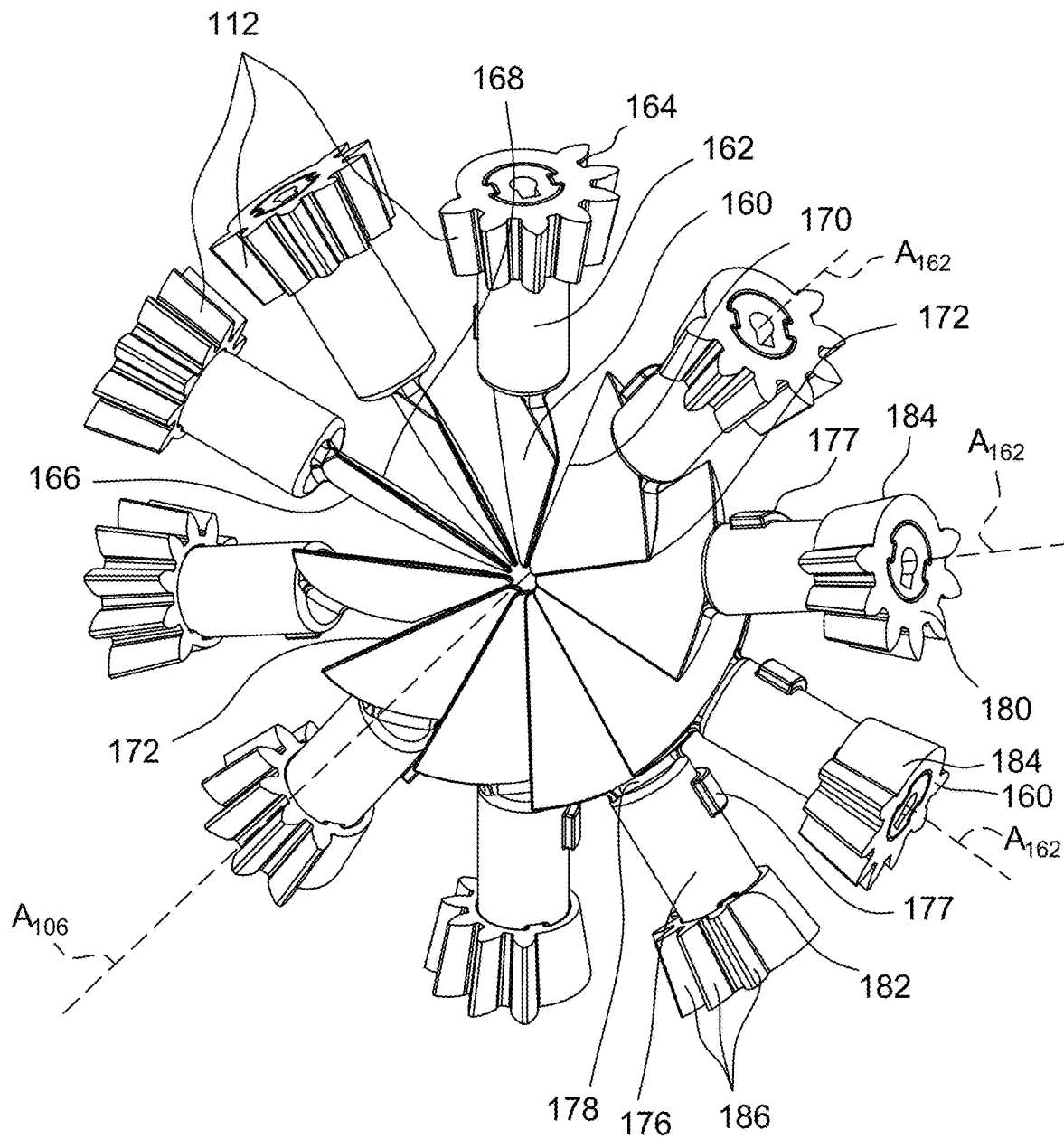


FIG. 7

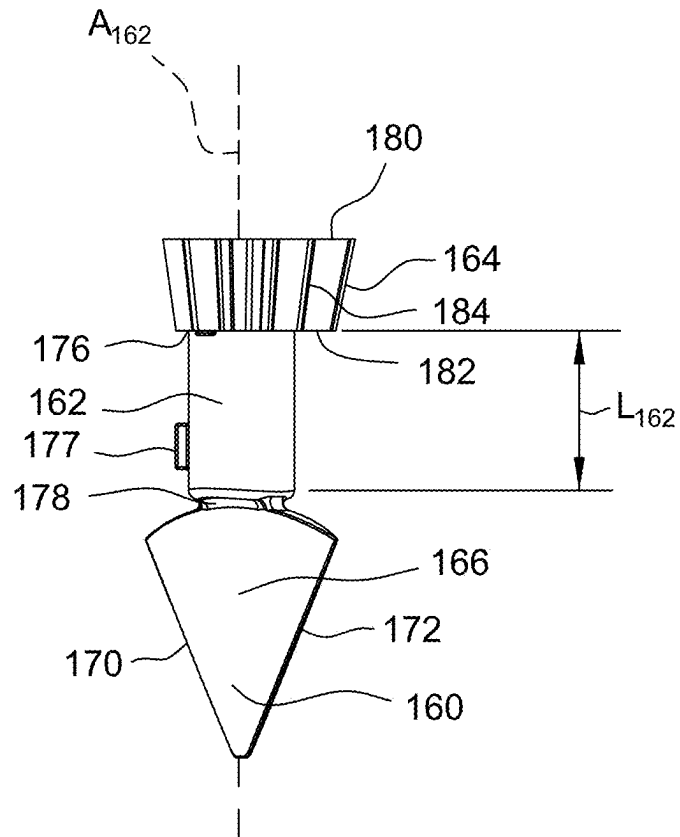


FIG. 8

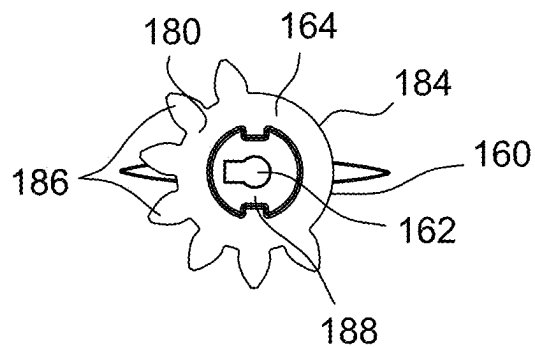


FIG. 9

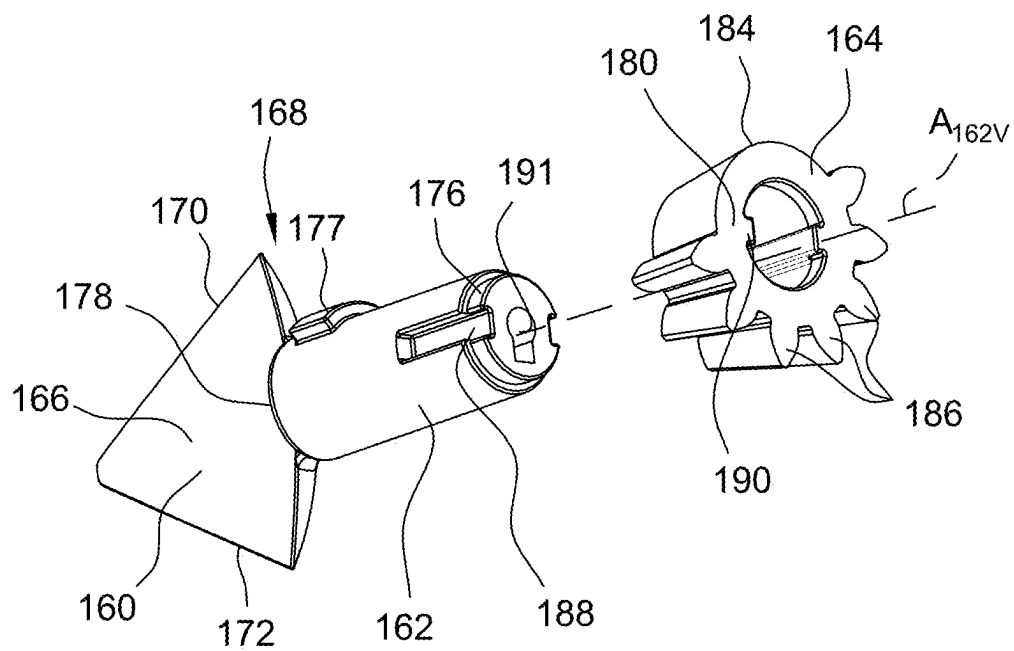


FIG. 10

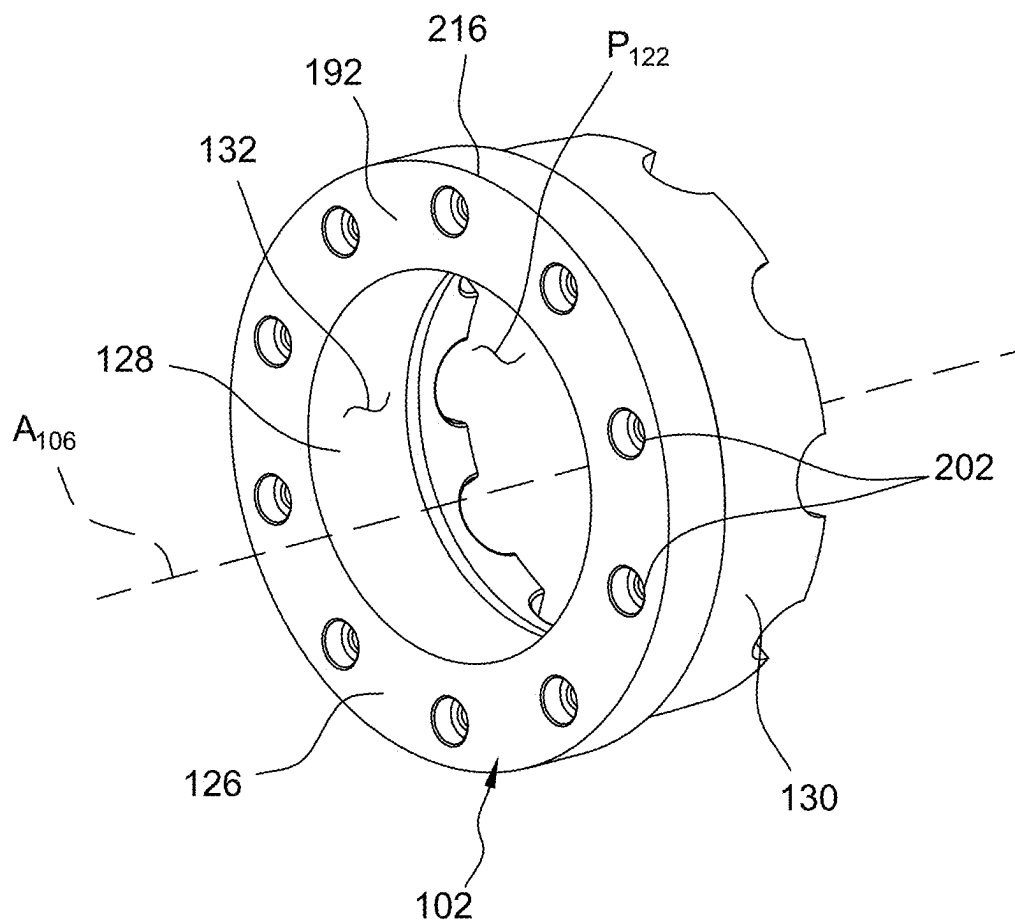


FIG. 11

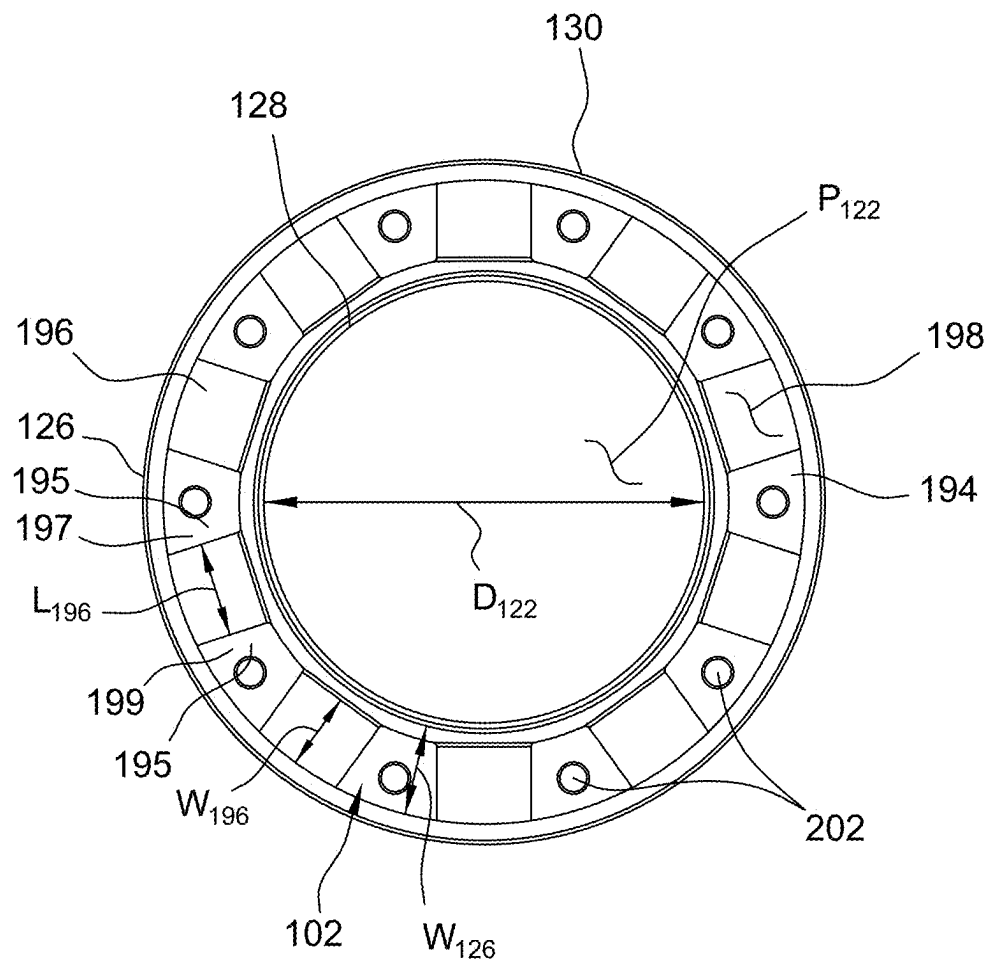


FIG. 12

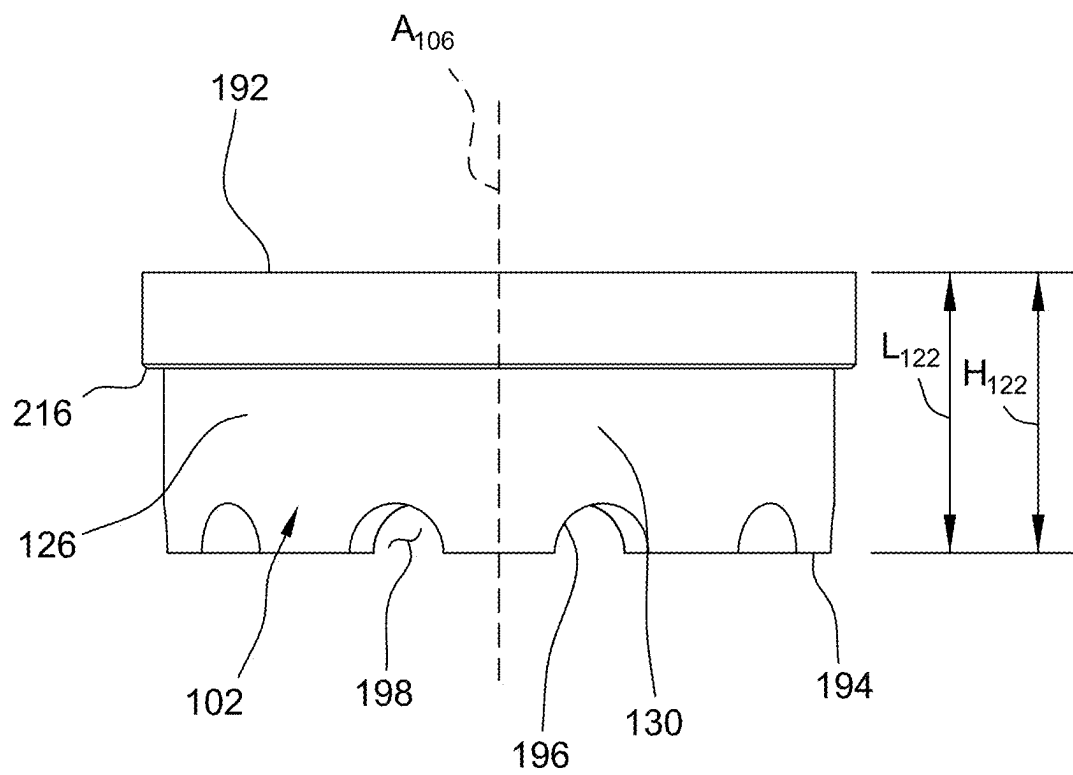


FIG. 13

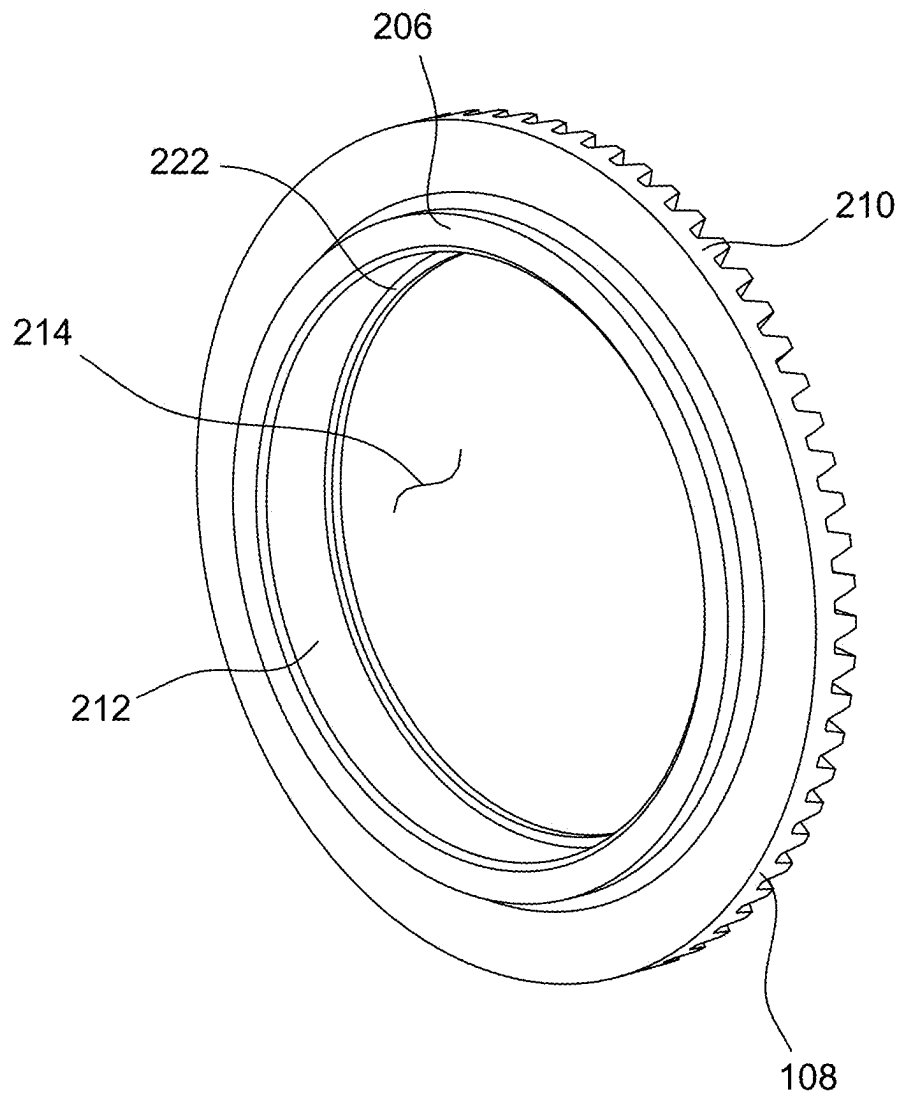


FIG. 14

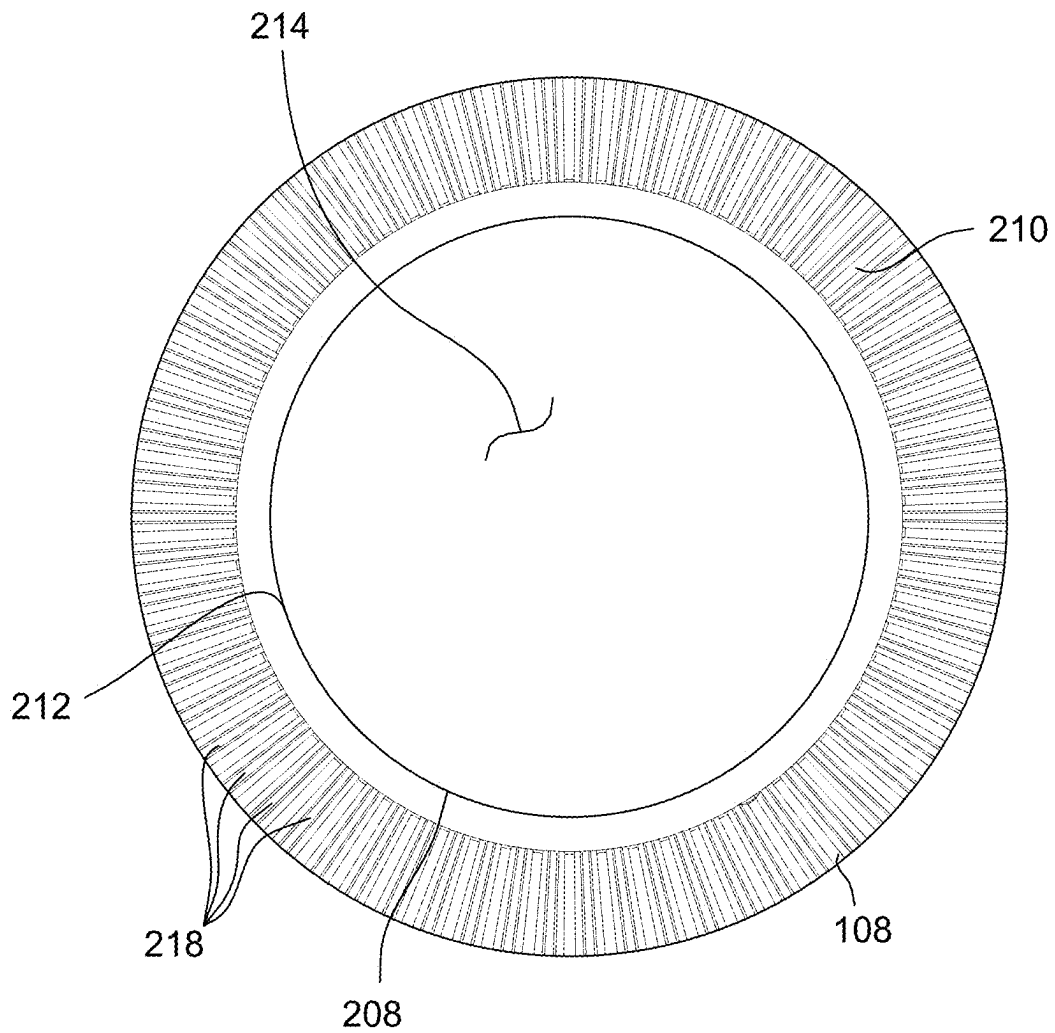


FIG. 15

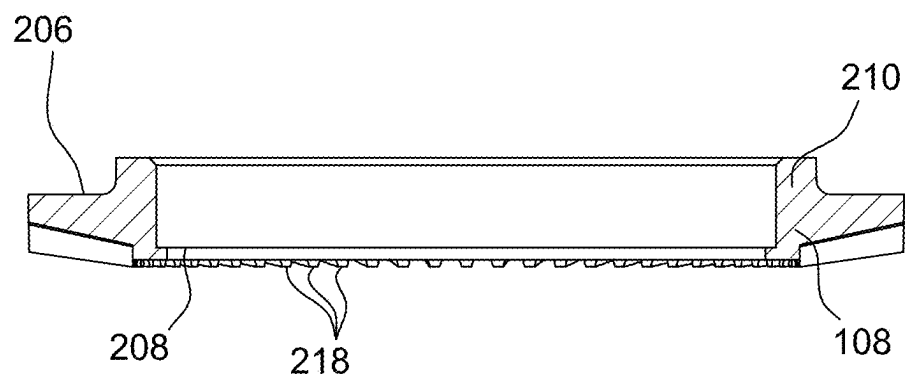


FIG. 16

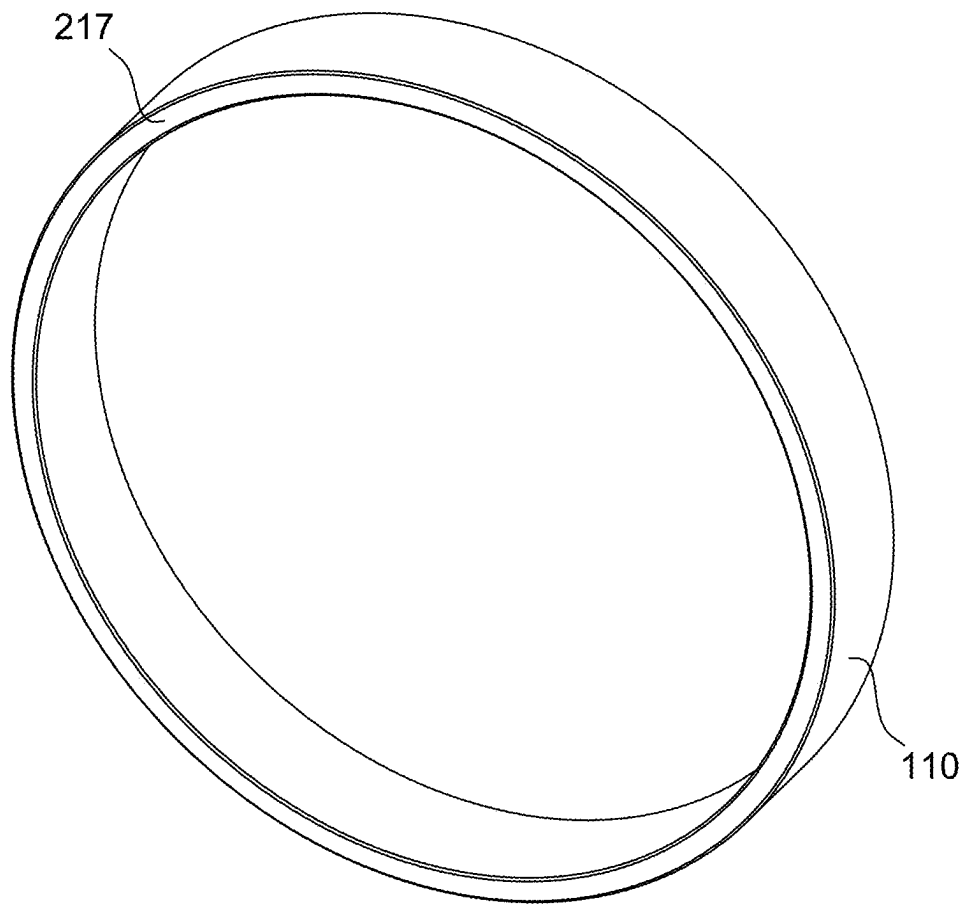


FIG. 17

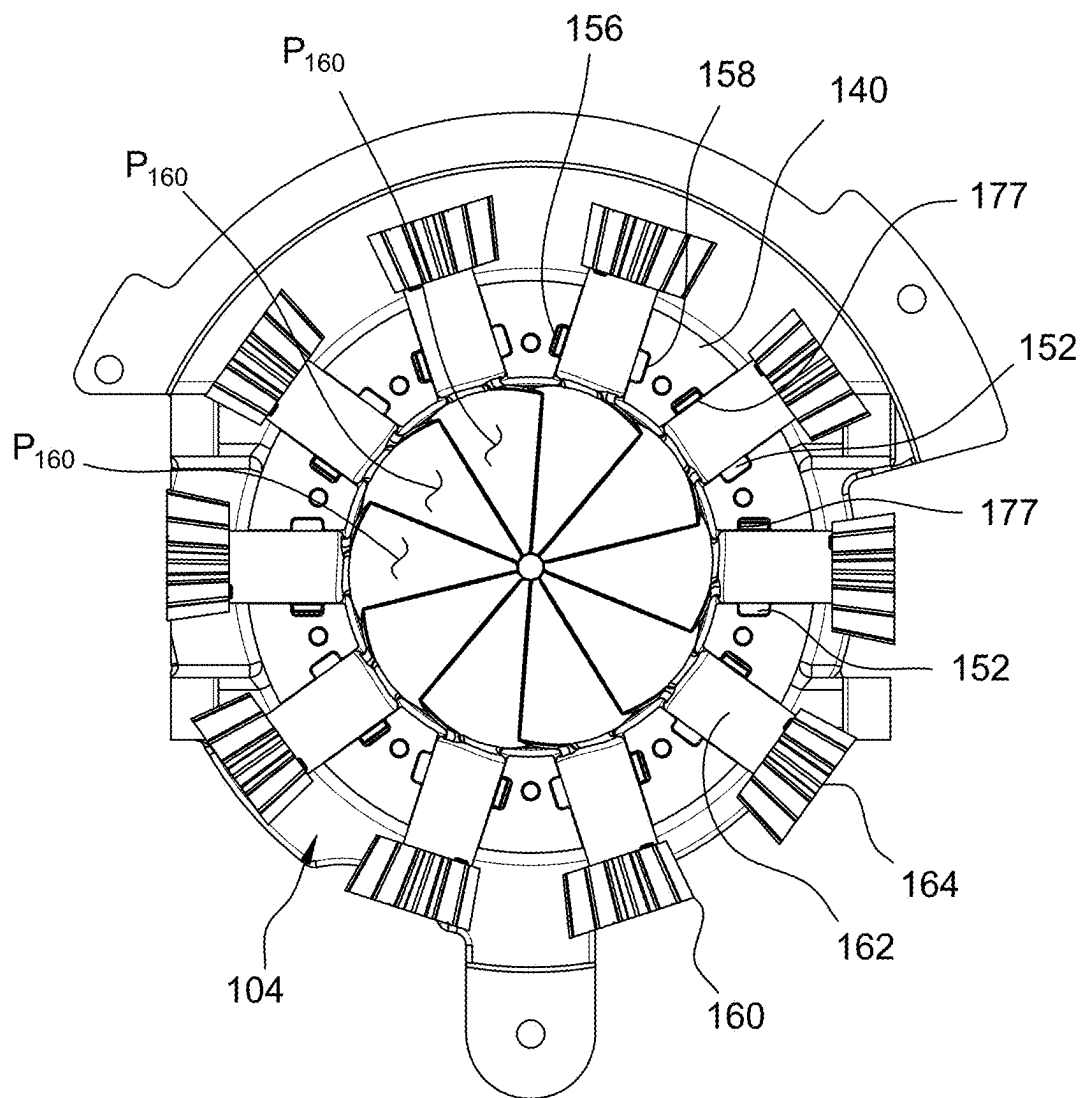
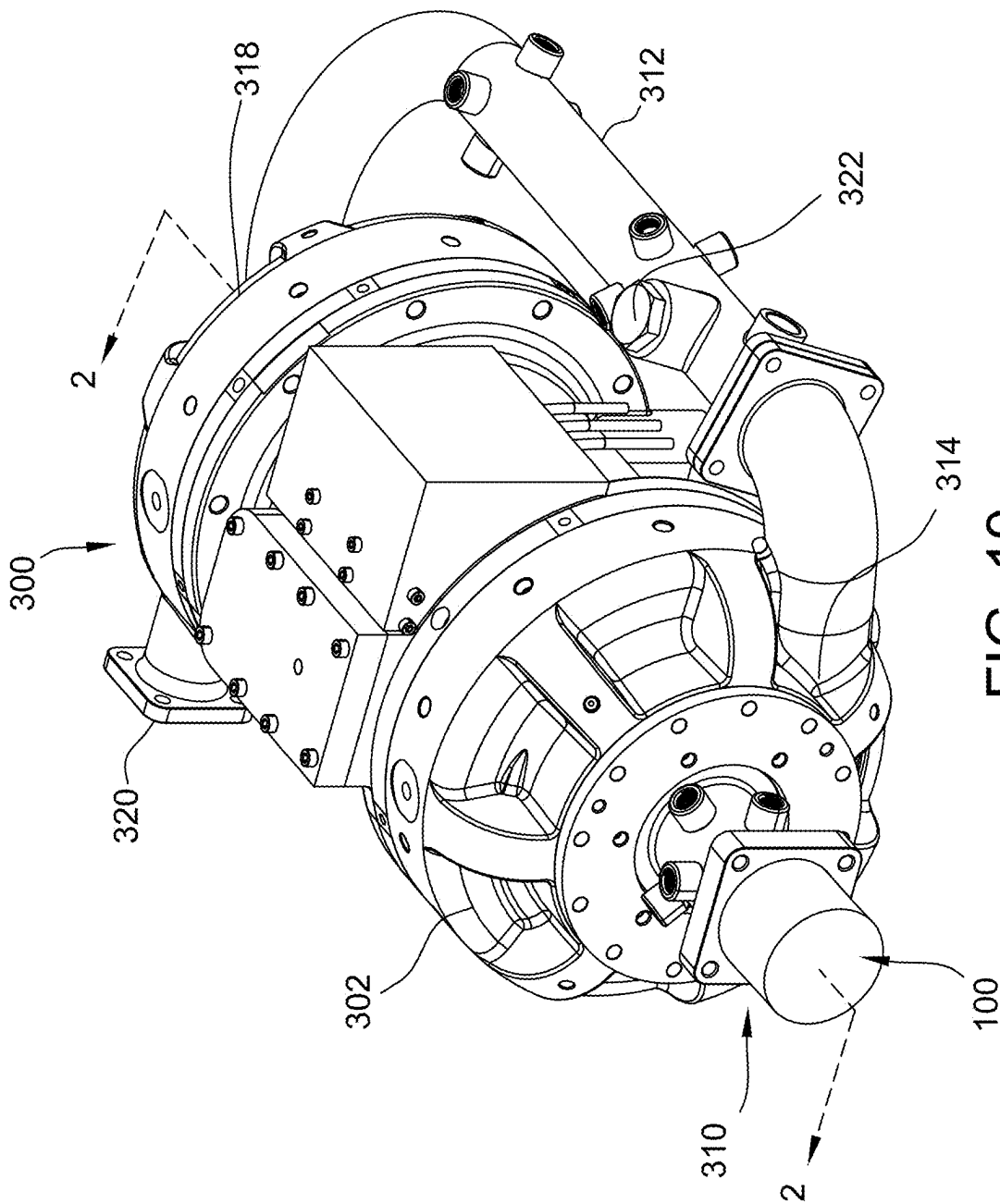


FIG. 18



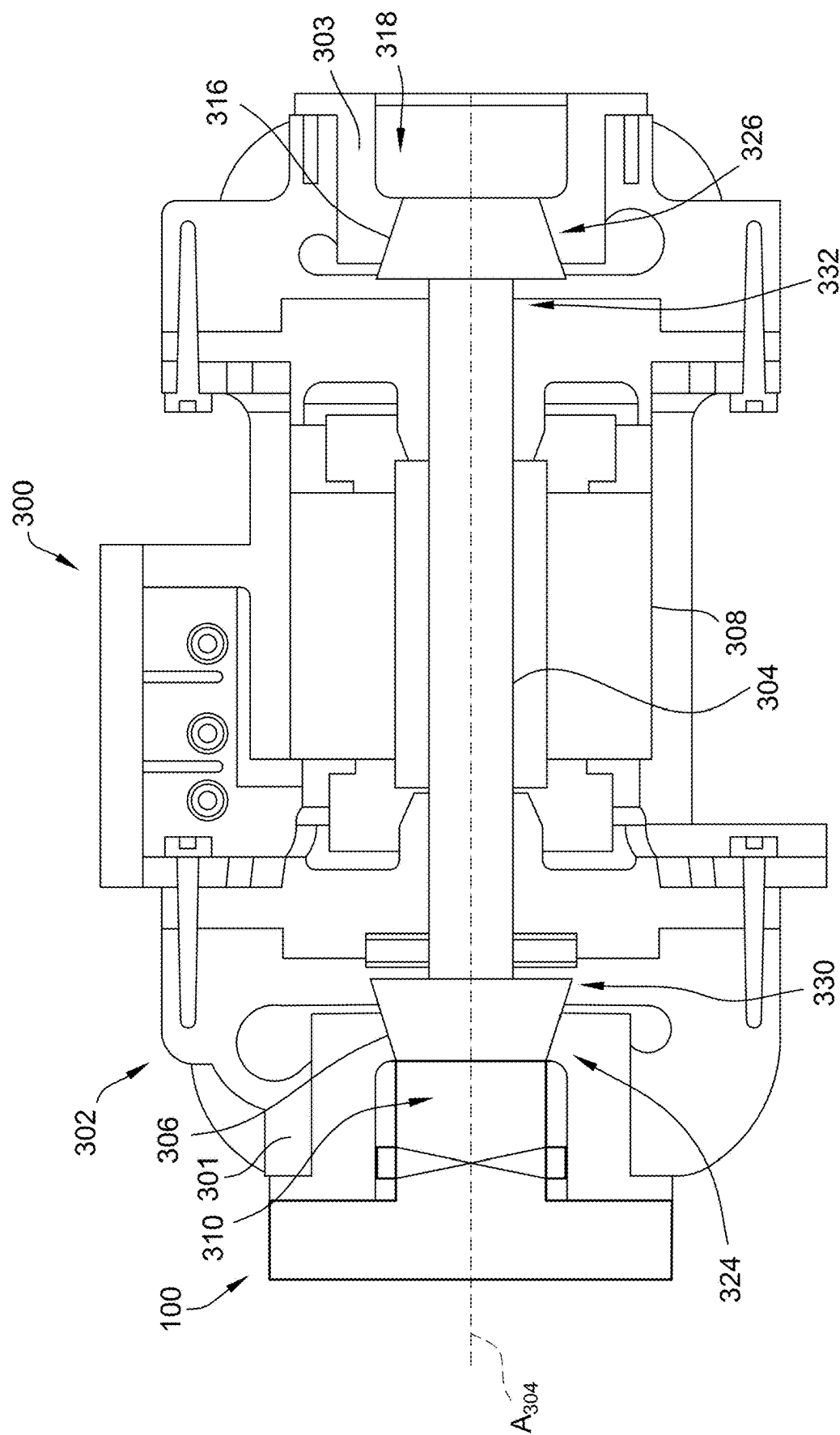


FIG. 20

1

VARIABLE INLET GUIDE VANE APPARATUS AND COMPRESSOR INCLUDING SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/656,260 filed on Mar. 24, 2022, entitled “Variable Inlet Guide Vane Apparatus and Compressor Including Same” hereby incorporated by reference.

FIELD OF THE DISCLOSURE

The field of the disclosure relates generally to compressors, and more particularly, to variable inlet guide vane apparatuses for use with compressors.

BACKGROUND

Inlet guide vane devices can be used to regulate pressure and direction of fluid flow at the inlet of a compressor, such as a centrifugal compressor. Conventional inlet guide vane devices include a plurality of guide vanes arranged circumferentially about a fluid flow path. The plurality of vanes impart a swirling motion to the fluid flow directing the fluid flow at suitable angles entering the compressor to improve efficiency and performance. The plurality of guide vanes may be rotated to adjust the orientation of the guide vanes relative to the fluid flow path to meet air intake requirements of the compressor for various operating conditions.

Each of the plurality of guide vanes may be rotatable relative to a housing mounted in proximity to the inlet of the compressor. High wear components of known inlet guide vane devices may be difficult to access for inspection and/or maintenance. Known inlet guide vane devices have one or more bearings which require maintenance, such as relubrication and/or periodic replacement, requiring downtime of the compressor. In some instances, the guide vane housing must be disconnected from the compressor, and then the guide vane housing disassembled during repair or inspection to access the guide vanes. The disassembly process can result in significant downtime of the compressor. Additionally, disassembly of the housing to access the guide vanes increases the likelihood of handling and installation damage, reducing the overall life of the inlet guide vane device. For example, disassembly of the housing may expose bearing surfaces to contaminants and debris potentially leading to etching and corrosion. Reassembling of the plurality of guide vanes relative to the housing may also result in misalignment.

This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the disclosure, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

SUMMARY

In one aspect an inlet guide vane apparatus includes a housing assembly defining a fluid flow passageway. The housing assembly includes a first housing portion and a second housing portion connected to and positioned axially upstream of the first housing portion. The first and second

2

housing portions cooperatively define a plurality of guide vane openings extending radially through the housing assembly and into the fluid flow passageway. A ring gear is rotatable relative to the housing assembly. A plurality of guide vanes is connected to the housing assembly. Each of the guide vanes includes a stem extending from a first end to a second end. At least a portion of the stem is disposed within one of the guide vane openings. A vane gear is disposed at the first end of the stem and operably connected to the ring gear. The vane gear is disposed at an exterior of the housing assembly and a vane is disposed at the second end of the stem and within the fluid flow passageway of the housing assembly. Each of the guide vanes is rotatable relative to the housing assembly such that an orientation of the vane within the fluid flow passageway is selectively adjustable.

In another aspect, a compressor includes a compressor housing including an inlet and a driveshaft rotatably supported within the compressor housing. The compressor further includes an impeller connected to the driveshaft and operable to impart kinetic energy to incoming refrigerant gas upon rotation of the driveshaft. An inlet guide vane apparatus is connected to the compressor inlet. The inlet guide vane apparatus includes a housing assembly defining a fluid flow passageway. The housing assembly includes a first housing portion and a second housing portion connected to and positioned axially upstream of the first housing portion. The first and second housing portions cooperatively define a plurality of guide vane openings extending radially through the housing assembly and into the fluid flow passageway. A ring gear is rotatable relative to the housing assembly and a plurality of guide vanes is connected to housing assembly. Each of the guide vanes includes a vane gear operably connected to the ring gear and disposed at an exterior of the housing assembly and a vane disposed within the fluid flow passageway of the housing assembly. Each of the guide vanes is rotatable relative to the housing assembly such that an orientation of the vane within the fluid flow passageway is selectively adjustable.

In yet another aspect, a method of assembling an inlet guide vane apparatus including a housing assembly and a plurality of guide vanes is provided. The housing assembly includes a first housing portion and a second housing portion. Each of the guide vanes includes a stem extending from a first end to a second end. A vane gear is disposed at the first end of the stem and a vane is disposed at the second end of the stem. The method includes rotatably connecting a ring gear to the first housing portion and connecting the first housing portion to the second housing portion such that the second housing portion is positioned axially upstream of the first housing portion, such that the first and second housing portions cooperatively define a plurality of guide vane openings extending radially through the housing assembly. The method further includes positioning the plurality of guide vanes relative to the second housing portion such that, when the first housing portion is connected to the second housing portion, at least a portion of the stem of each guide vane is disposed within one of the guide vane openings and the vane gear of each guide vane is positioned at an exterior of the housing assembly. The method includes operably connecting the vane gear of each guide vane to the ring gear.

Various refinements exist of the features noted in relation to the above-mentioned aspects of the present disclosure. Further features may also be incorporated in the above-mentioned aspects of the present disclosure as well. These refinements and additional features may exist individually or

3

in any combination. For instance, various features discussed below in relation to any of the illustrated embodiments of the present disclosure may be incorporated into any of the above-described aspects of the present disclosure, alone or in any combination.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of an example a variable inlet guide vane apparatus including a first and second housing portion, a bearing, a ring gear, and a plurality of guide vanes.

FIG. 2 is a side view of the variable inlet guide vane apparatus shown in FIG. 1.

FIG. 3 is an exploded view of the variable inlet guide vane apparatus shown in FIG. 1.

FIG. 4 is a perspective view of the second housing portion for use with the variable inlet guide vane apparatus shown in FIG. 1.

FIG. 5 is a rear or downstream view of the second housing portion shown in FIG. 4.

FIG. 6 is a sectional view of the second housing portion shown in FIG. 4.

FIG. 7 is a perspective view of the plurality of guide vanes for use with the variable inlet guide vane apparatus shown in FIG. 1.

FIG. 8 is a side view of one of the plurality of guide vanes shown in FIG. 7.

FIG. 9 is a top view of the guide vane shown in FIG. 8.

FIG. 10 is an exploded view of the guide vane shown in FIG. 8.

FIG. 11 is a perspective view of the first housing portion for use with the variable inlet guide vane apparatus shown in FIG. 1.

FIG. 12 is a front or upstream view of the first housing portion shown in FIG. 11.

FIG. 13 is a side view of the first housing portion shown in FIG. 11.

FIG. 14 is a perspective view of the ring gear for use with the variable inlet guide vane apparatus shown in FIG. 1.

FIG. 15 is an end view of the ring gear shown in FIG. 14.

FIG. 16 is a sectional view of the ring gear shown in FIG. 14.

FIG. 17 is a perspective view of the bearing of the variable inlet guide vane apparatus shown in FIG. 1.

FIG. 18 is a rear or downstream view of the variable inlet guide vane apparatus having the first housing portion, ring gear, and bearing disconnected, illustrating the plurality of guide vanes arranged within a second channel on the second housing portion.

FIG. 19 is a perspective view of an assembled compressor for use with the variable inlet guide vane apparatus.

FIG. 20 is a cross-sectional view of the compressor of FIG. 19 taken along line 2-2.

Corresponding reference characters indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

FIG. 1 is a perspective view of an example variable inlet guide vane apparatus, generally indicated at 100. The variable inlet guide vane apparatus 100 (also referred to herein as the inlet guide 100) is suitable for use with compressors, such as centrifugal compressors (see, e.g., compressor 300, shown in FIGS. 19 and 20), and can improve the operating range and efficiency of the compressor by imparting a pre-swirl motion to a fluid flow F entering the compressor. The inlet guide 100 may be mounted in proximity to an inlet

4

of the compressor and the fluid flow F exits the inlet guide 100 with a pre-swirl and enters into the inlet of the compressor, such that the fluid flow F contacts an impeller of the compressor with a suitable direction. Alternatively, and/or additionally, the inlet guide vane 100 is mounted in proximity to the inlet for each stage of a multi-stage compressor.

FIG. 2 is a side view of the inlet guide 100, and FIG. 3 is an exploded view of the inlet guide 100. In the illustrated embodiment, the inlet guide 100 includes a first housing portion 102 and a second housing portion 104. The first housing portion 102 and the second housing portion 104 can be connected to form a vane housing assembly 106. The second housing portion 104 is positioned axially upstream from the first housing portion 102 relative to the direction of fluid flow. The inlet guide 100 also includes a ring gear 108 rotatably connected to the housing assembly 106. In the illustrated embodiment, the ring gear 108 is rotatably connected to the first housing portion 102. The ring gear 108 can be rotatably connected to the housing assembly 106 by a bearing, such as bearing 110 (FIG. 3). In other embodiments, the bearing 110 may be omitted.

Inlet guide 100 further includes a plurality of guide vanes 112. Each guide vane 112 is rotatable relative to the vane housing assembly 106 and is operably connected with the ring gear 108 such that rotation of the ring gear 108 causes each of the plurality of guide vanes 112 to rotate in unison. Each of the guide vanes 112 is rotatable relative to the housing assembly 106 such that the orientation of the vane 160 within a fluid flow passageway P of the housing assembly 106 is selectively adjustable. In some embodiments, the guide vanes 112 are rotatable relative to the housing assembly 106 in unison.

The inlet guide 100 can also include one or more motors 174 operably connected to one or more of the guide vanes 112 to selectively rotate the guide vanes 112. The illustrated inlet guide 100 includes a motor 174 mounted to a motor mount on the second housing portion 104. The motor 174 is operably coupled to one of the guide vanes 112, also referred to as a drive guide vane, by a driveshaft 175. As described further herein, rotation of the drive guide vane by the motor 174 causes rotation of the ring gear 108, which in turn causes rotation of the other guide vanes 112, also referred to as follower guide vanes. The motor 174 can include any suitable motor that enables the inlet guide 100 to function as described herein, including, for example and without limitation, electric motors. In some embodiments, the motor 174 is a stepper motor. In other embodiments, the motor 174 includes a synchronous motor that operates based on rotation feedback from one or more of the follower vanes, e.g., using a potentiometer or position sensor coupled to one or more of the follower vanes, to ensure all vanes are driven to the desired angle. The motor 174 may include for example and without limitation a DC motor, brushed or brushless, and a synchronous AC motor.

With additional reference to FIG. 3, the first housing portion 102 includes a first annular wall 126 having a first inner surface 128 and a first outer surface 130. The second housing portion 104 includes a second annular wall 120 having a second inner surface 122 and a second outer surface 124. The first inner surface 128 and the second inner surface 122 defines the boundary of a fluid flow passageway P extending through the vane housing assembly 106. The vane housing assembly 106 has a housing axis A_{106} extending through the fluid flow passageway P. The first housing portion 102 defines an exit or outlet 132 of fluid flow passageway P, and the second housing portion 104 defines an inlet 134 of the fluid flow passageway P. The fluid flow

5

F enters the vane housing assembly **106** at the inlet **134**, passes through the fluid flow passageway P, and leaves the vane housing assembly **106** at the outlet **132**. The fluid flow F flows through the fluid flow passageway P in a direction that is generally parallel to the housing axis A_{106} . Fluid flow F leaving the exit **132** has a pre-swirl imparted by the plurality of the guide vanes **112**, as described in further detail herein.

The vane housing assembly **106** includes an exterior area **136** surrounding the first outer surface **130** and the second outer surface **124** and located generally radially outward from the housing assembly **106**. In the illustrated embodiment, at least a portion of each of the plurality of guide vanes **112** is disposed between the first housing portion **102** and second housing portion **104**, and at least a portion of the guide vane **112** and the ring gear **108** is arranged in the exterior area **136** of the vane housing assembly **106**. Accordingly, the ring gear **108** and at least a portion of the guide vane **112** are accessible (e.g., to an operator or technician) for inspection and/or repairs without requiring inlet guide **100** to be disassembled. By way of example, an operator or technician can access the ring gear **108** and a portion of the guide vanes **112** (e.g., vane gears, described in more herein) without first disconnecting the first housing portion **102** from the second housing portion **104**.

FIG. 4 is a perspective view of the second housing portion **104**. FIGS. 5 and 6 are a rear view and a sectional view, respectively, of the second housing portion **104**. The second annular wall **120** includes a downstream surface **140** and an upstream surface **142**. The downstream surface **140** is generally annular in shape. The second annular wall **120** may have a width W_{120} extending between the second outer surface **124** and the second inner surface **122**. (FIG. 5). The second annular wall **120** may have a height H_{120} extending between the downstream surface **140** and the upstream surface **142**. (FIG. 6). The second inner surface **122** defines the boundary of a second fluid flow passageway P_{120} that includes a diameter D_{120} defined by the second inner surface **122**. In the illustrated embodiment, the second fluid flow passageway P_{120} is generally conical in shape and the diameter D_{120} decreases in a direction from the inlet **134** to the outlet **132** along the axis A_{106} . In other embodiments, the second fluid flow passageway P_{120} can be generally cylindrical in shape and the diameter D_{120} defined by the second inner surface **122** is generally constant. The second fluid flow passageway P_{120} has a length L_{120} . In the illustrated embodiment, the length L_{120} corresponds to the height H_{120} of the second annular wall **120**. The housing axis A_{106} extends through the second fluid flow passageway P_{120} . The upstream surface **142** is generally planar and may be mounted to a compressor. Alternatively, the second housing portion **104** may be mounted to any suitable structure in proximity to the inlet of a compressor. The dimensions of the compressor, e.g., width W_{120} , height H_{120} , diameter D_{120} , and length L_{120} , may be scaled to the size of the compressor and the aerodynamic needs of the compressor.

With reference to FIGS. 4-6, the second housing portion **104** includes one or more flanges **144** extending radially outward from the second annular wall **120**. The flange **144** may extend generally perpendicular to the second annular wall **120**. The flanges **144** include one or more fastener openings **146** for receiving suitable fasteners (e.g., screws, bolts, etc.) to connect the second housing portion **104** to a compressor.

With reference again to FIG. 5, the downstream surface **140** includes a plurality of second channel surfaces **148**, each defining a corresponding second channel **150**. Each of

6

the plurality of second channel surfaces **148** are arranged in a radially symmetric pattern about the housing axis A_{106} . In the illustrated embodiment, the downstream surface **140** of the second annular wall **120** includes ten second channel surfaces **148** defining ten second channels **150** arranged in a radially symmetric pattern about the housing axis A_{106} . In alternative embodiments, the downstream surface **140** may include any number of second channel surfaces **148** that enables the inlet guide **100** to function as described herein. For example, in some embodiments, there are six second channel surfaces **148** defining six second channels **150**.

In the illustrated embodiment, each of the plurality of second channel surfaces **148** are identical, having the same size and shape. In the illustrated embodiment, the second channel surfaces **148** are in the shape of a segment of a cylindrical surface. Accordingly, the second channels **150** are generally in the shape of a half-cylinder. The second channel surface **148** has a second channel length L_{148} extending from the second inner surface **122** to the second outer surface **124**. In the illustrated embodiment, the second channel surfaces **148** and second channels **150** extend through the entire width W_{120} of the second annular wall **120**. In other embodiments, the second channel surfaces **148** extend only partially through the width W_{120} of the second annular wall **120**. The second channel surfaces **148** are sized and shaped such that second channels **150** are sized and shaped to receive at least a portion of the guide vanes **112** therein, as described in further detail herein.

In the illustrated embodiment, each second channel surface **148** includes a secondary channel surface **152** defining a slot **154**. The secondary channel surface **152** extends radially from the second channel surface **148**, such that the slot **154** has a depth D_{152} extending from the second channel surface **148** (FIG. 4). The secondary channel surface **152** defining the slot **154** includes a first end **156** and a second end **158**, and a secondary channel length L_{152} extending therebetween (FIG. 5). The secondary channel surface further defines a slot width W_{152} .

FIG. 7 is a perspective view of the plurality of guide vanes **112**. Each of the plurality of guide vanes **112** includes a vane **160**, a stem **162**, and a vane gear **164**. The plurality of guide vanes **112** are arranged in a radially symmetric pattern mirroring the radially symmetric pattern of the plurality of second channel surfaces **148** on the second housing portion **104**. In the illustrated embodiment, there are ten guide vanes **112** corresponding to the ten second channels surfaces **148**. In other embodiments, the inlet guide **100** can include any suitable number of guide vanes **112** that enables the inlet guide **100** to function as described herein. For example, in some embodiments there may be six guide vanes **112** corresponding to six second channel surfaces **148**.

In the illustrated embodiment, the vane **160** is formed integrally with the stem **162**. For example, the vane **160** and the stem **162** may be molded as a single piece. In alternative embodiments, the vane **160** may be formed separately from and connected to the stem **162**. The vane **160** is substantially triangular, in the illustrated embodiment, and includes a first vane side **166** and an opposing second vane side **168**. The first and second vane sides **166**, **168** are substantially planar. The first and second vane sides **166** and **168** connect at a trailing edge **170** and a leading edge **172**. The trailing and leading edges **170** and **172** may be knife-like in shape. The plurality of vanes **160** are selectively rotatable to selectively occlude or block, fluid flow F through the fluid flow passageway P_{120} . The plurality of vanes **160** may be prevented from being rotated to arrange the vanes **160** in a closed position, in which the trailing edge **170** of each guide vane

is in contact with, or in close proximity with, the leading edge 172 of an adjacent guide vane, as described herein.

The first and second housing portions 102 and 104 cooperatively define a plurality of guide vane passages P_{160} . Each of the plurality of vanes 160 may cover one of the vane passages P_{160} . Each vane passage P_{160} is a portion of the fluid flow passageway P. The vane passages P_{160} are spaced circumferentially about the fluid flow passageway P. The vane 160 is any shape or size enabling the inlet guide 100 to function as describe herein. Additionally, the shape and size of the vane 160 may be selected based upon the intended application of the inlet guide 100. For example, the size, shape, and angle of the vane 160 may be selected based on the type and configuration of the compressor, the operating conditions, and/or the fluid type used with the compressor. Each of the guide vanes 112 is rotatable relative to the vane housing assembly 106 such that the orientation of the vane 160 within the fluid flow passageway P is selectively adjustable.

FIGS. 8 and 9 are side and top views, respectively, of one of the guide vanes 112 shown in FIG. 7. The stem 162 extends along a stem axis A_{162} from a first, inner end 178 to a second, outer end 176. The vane 160 is disposed at the first, inner end 178 of the stem 162, and the vane gear 164 is disposed at the second, outer end 176 of the stem 162. The stem 162 has a stem length L_{162} between the inner end 178 and the outer end 176. The stem length L_{162} may be substantially similar to the second channel length L_{148} . When the stem 162 is arranged within one of the plurality of second channels 150, the stem axis A_{162} is perpendicular to the housing axis A_{160} .

In this illustrated embodiment, the guide vane 112 includes a stop 177 extending circumferentially about the stem 162, and radially outward in a direction generally perpendicular to the stem axis A_{106} , from the stem 162. The stop 177 is sized and shaped to fit within the slot 154 defined by the secondary channel surface 152. In the illustrated embodiment, the stop 177 is generally rectangular in shape. The stop 177 and the secondary channel surface 152 are sized and shaped such that the stop 177 slides along the length L_{152} relative to the secondary channel surface 152 when the guide vane 112 rotates relative to the first housing portion 102 and the second housing portion 104 about the stem axis A_{162} . In addition, the slot width W_{152} is sized and shaped such that the stop 177 contacts the secondary channel surface 152, preventing the guide vane 112 from translating along a direction parallel to the stem axis A_{162} . The slot width W_{152} is sized to provide sufficient clearance between the stop 177 and the secondary channel surface 152 such that the stop 177 may translate along the slot length L_{152} .

The stop 177 extends only partially around the circumference of the stem 162. In the illustrated embodiment, the stop 177 extends circumferentially an arc angle of about 45° around the circumference of the stem 162. In other embodiments, the stop 177 may extend an arc angle greater than or less than 45°. For example, in some embodiments, the stop 177 may extend an arc angle of 90° around the stem 162. In another example, the stop 177 may extend an arc angle of 30° around the stem 162.

The stop 177 limits rotation of the guide vane 112 about the stem axis A_{162} when the stop 177 engages with a stop surface of at least one of the first housing portion 102 or the second housing portion 104, as described further herein.

The vane gear 164 includes a top surface 180, a bottom surface 182, and a wall 184 extending between the top surface 180 and the bottom surface 182. The wall 184 is generally cylindrical in shape (FIG. 8). The vane gear 164

includes a plurality of gear teeth 186 extending radially outward from the wall 184. In some embodiments, such as the illustrated embodiment, the vane gear 164 is a tapered gear—i.e., the gear teeth 186 are tapered or angled radially inward or outward from the top surface 180 to the bottom surface 182. In some embodiments, the vane gear 164 may be a bevel gear. In some embodiments, the vane gear 164 may be a helical gear.

In the illustrated embodiment, the vane gear 164 is a partial gear in which the gear teeth 186 extend around only a portion of the wall 184. In the illustrated embodiment, the gear teeth 186 extend an arc angle of approximately 225° around the wall 184. In other embodiments, the gear teeth 186 may extend around the wall 184 an arc angle of greater than or less than 225°. Further, in the illustrated embodiment, the gear teeth 186 are positioned generally opposite the stop 177 on the stem 162. Accordingly, during operation, the vane gear 164 is arranged to engage with the ring gear 108, while the stop 177 is captured within the slot 154.

FIG. 10 is an exploded view of the guide vane 112. In some embodiments, the vane gear 164 is removably connected to the stem 162 of the guide vane 112. In the illustrated embodiment, for example, the outer end 176 of the stem 162 is received within a central opening of the vane gear 164 to connect the vane gear 164 to the stem 162. In some embodiments, the stem 162 and the vane gear 164 may be connected using a press-fit engagement when the outer end 176 is disposed within the central opening of the vane gear 164. Additionally, and/or alternatively, an epoxy, or other suitable adhesive, may be used to connect the stem 162 and the vane gear 164. Additionally, in the illustrated embodiment, the stem 162 includes a key 188 disposed at the outer end 176 and the vane gear 164 includes a keyed boundary 190 defining a keyed opening 189 sized and shaped to receive the key 188. When the key 188 is disposed within the keyed opening 189, the stem 162 and the vane gear 164 are frictionally engaged and rotation of the vane gear 164 is transmitted to the stem 162. In some embodiments, the key 188 may be press fit within the keyed opening 189. In some embodiments, the key 188 may include a channel, and the boundary 190 may include a key that is sized and shaped to fit within the channel. Alternatively, and/or additionally, the key 188 and the boundary 190 may include any suitable features that enable frictional engagement of the vane gear 164 and the stem 162. In other embodiments, the vane gear 164 is integral with the guide vane (e.g., the vane gear 164 is formed integrally with the stem 162).

In the illustrated embodiment, the guide vane 112 includes an alignment feature 191. The alignment feature 191 receives a portion of the driveshaft of a motor, such as driveshaft 175 of motor 174, enabling the guide vane 112 to be operably connected to the motor. Accordingly, the alignment feature 191 is shaped complimentary to the driveshaft. For example, the alignment feature 191 may be keyed, semicircular, or star shaped, or any suitable shape such that the alignment feature 191 mates with the driveshaft in order to frictionally couple the driveshaft to the alignment feature 191. Alternatively, and/or additionally, the alignment feature 191 may be sized and shaped to receive an alignment tool (not shown) to facilitate alignment and mounting of the guide vanes 112 to the vane housing assembly 106.

FIG. 11 is a perspective view of the first housing portion 102. The first annular wall 126 includes a downstream surface 192 and an upstream surface 194. The upstream surface 194 is generally annular in shape. The first annular wall 126 has a width W_{126} extending between the first outer

surface **130** and the first inner surface **128**. (FIG. **12**). The first annular wall **126** may have a height H_{126} of extending between the downstream surface **192** and the upstream surface **194** (FIG. **13**). The first inner surface **128** defines the boundary of a first fluid flow passageway P_{122} . The first fluid flow passageway P_{122} is generally cylindrical in shape having a diameter D_{122} defined by the first inner surface **128**. In the illustrated embodiment, the diameter D_{122} is substantially similar to the diameter D_{120} of the second inner surface **122**. (FIG. **12**) The first fluid flow passageway P_{122} has a length L_{122} corresponding the height H_{122} of the first annular wall **126** (FIG. **13**). The housing axis A_{160} extends through the first fluid flow passageway P_{122} . When the first housing portion **102** and the second housing portion **104** are connected, creating the vane housing assembly **106**, the first fluid flow passageway P_{120} and the second fluid flow passageway P_{122} are aligned creating the fluid flow passageway P . The dimensions of the compressor, e.g., width W_{126} , height H_{126} , diameter D_{122} , and length L_{122} , may scaled to the size of the compressor and/or the aerodynamic needs of the compressor.

FIG. **12** is a front view of the first housing portion **102**. The upstream surface **194** of the first housing portion **102** includes a plurality of first channel surfaces **196** that define a plurality of first channels **198**. The plurality of first channel surfaces **196** are arranged in a radially symmetric pattern, about the housing axis A_{106} that mirrors the radially symmetric pattern of the second channel surfaces **148** and the radially symmetric pattern of the plurality of guide vanes **112**. In the illustrated embodiment, each of the first channel surfaces **196** are substantially similar having approximately the same size and shape. In other example embodiments, the first channel surfaces **196** are identical to the second channel surfaces **148**. Each of the first channel surfaces **196** includes a first channel length L_{196} and a first channel width W_{196} (FIG. **12**). The first channel width W_{196} may extend from the first inner surface **128** to the first outer surface **130**.

In the illustrated embodiment, the first channel surfaces **196** are in the shape of a segment of a cylindrical surface. Accordingly, the first channels **198** are generally half-cylindrical in shape. In the illustrated embodiment, the first channel surface **196** extends along the entire width W_{126} of the upstream surface **194**. In other embodiments, the first channel surfaces **196** extend only partially along the width of the upstream surface **194**.

As mentioned above, the stop **177** limits rotation of the guide vanes **112** about the stem axis A_{162} when the stop **177** engages one or more stop surfaces **195**. In the illustrated embodiment, for example, the first housing portion **102** includes a first stop surface **197** and a second stop surface **199** (FIG. **12**). When the guide vane **112** rotates in a first direction (e.g., counterclockwise), the stop **177** slides within slot **154**, until the stop **177** engages the first stop surface **197**, preventing or inhibiting further rotation of the guide vane **112**. When the guide vane **112** rotates in a second direction (e.g., clockwise), the stop **177** slides within slot **154**, until the stop **177** engages the second stop surface **199**, preventing or inhibiting further rotation of the guide vane **112**.

In some embodiments, contact between the stop **177** and the stop surfaces **195** serves as a stop for a stepper motor. Specifically, in the illustrated embodiment, the stop **177** is disposed within the slot **154** and when the guide vane **112** rotates, the stop **177** travels along the slot length L_{154} . The guide vanes **112** may include a drive guide vane operably connected to a motor, and a plurality of follower guide vanes that rotate in response to rotation of the drive guide vane. In such embodiments, the stepper motor may rotate the drive

guide vane **112** until the stop **177** on one of the guide vanes **112** engages with either the first stop surface **197** or the second stop surface **199**, arresting the rotational motion of all of the guide vanes **112** and stopping the stepper motor.

In the illustrated embodiment, the slot **154**, the stop **177**, and the first stop surface **197** and the second stop surface **199** are configured to allow the motor to rotate the guide vanes **112** a total of 90° . For example, the plurality of guide vanes **112** are positionable in a first rotational position of 0° , or a neutral position, such that the first vane side **166** or the second vane side **168** of each guide vane **112** is arranged generally parallel to the housing axis A_{106} and/or parallel to the fluid flow F entering the inlet **134**. In the illustrated embodiment, when the guide vanes **112** are in the neutral position, the stop **177** is arranged in a center of the slot **154**. The stop **177** and the stop surfaces **195** are arranged to allow the plurality of guide vanes **112** to be positionable between $+45^\circ$ and -45° degrees relative to the neutral position such that the first vane side **166** or the second vane side **168** may be arranged at a desired angle relative to the housing axis A_{106} and the fluid flow F entering the inlet **134** depending on the desired operating condition.

In some embodiments, the second housing portion **104** can include one or more of the stop surfaces **195**. For example, in some embodiments, the first end **156** and the second end **158** includes stop surfaces that interact with stop **177**. In such embodiments, the guide vane **112** rotates, in either the first or second directions, until the stop **177** engages with the stop surfaces, preventing or inhibiting further rotation of the guide vane **112**.

When the first housing portion **102** is connected to the second housing portion **104**, each of the second channels **150** are aligned with each of the first channels **198**, such that the first and second channels **198** and **150** cooperatively form a plurality of guide vane openings **200** extending radially through the vane housing assembly **106** (FIG. **2**). The boundary of the guide vane openings **200** is defined by the first channel surface **196** and second channel surface **148**. Each guide vane opening **200** is generally cylindrical and is sized and shaped to receive at least a portion of the stem **162** of one of the guide vanes **112** therein. The stem **162** of each guide vane **112** is rotatable relative to the first channel surface **196** and second channel surface **148** such that each guide vane **112** rotates within one of the guide vane openings **200** about its respective stem axis A_{162} . In some embodiments, the first channel surface **196** and the second channel surface **148** includes a plain bearing to facilitate rotation of the stem **162** relative to the first channel surface **196** and the second channel surface **148**. Additionally, and/or alternatively, the stem **162** and the first channel surface **196** and second channel surface **148** may include suitable bearings enabling the inlet guide **100** to function as described herein. In some embodiments, the stem **162** and/or the first channel surface **196** and the second channel surface **148** may be impregnated with Teflon or other suitable lubricants. In the example embodiment, the inlet guide vane **100** may be used with an oil free compressor.

The first housing portion **102** and the second housing portion **104** may be connected in any suitable manner that enables the inlet guide **100** to function as described herein. In the illustrated embodiment, the first housing portion **102** is connected to the second housing portion **104** by a plurality of screws. More specifically, the first housing portion **102** includes a plurality of fastener openings **202** extending through the first annular wall **126**, and the second housing portion **104** includes a plurality of fastener opening **204** corresponding to the arrangement of the plurality of open-

11

ings 202 on the first housing portion 102. Each of the fastener openings 204 may include a threaded portion, such that a threaded bolt or screw (not shown) may be inserted through the opening 202 on the first housing portion 102 and threaded into the threaded portion of the openings 204 on the second housing portion 104. In other embodiments, the first housing portion 102 and second housing portion 104 may be connected using any suitable fasteners to connect the first housing portion 102 and second housing portion 104 together forming the vane housing assembly 106.

When the stem 162 of each guide vane 112 is arranged within one of the guide vane openings 200 of the vane housing assembly 106, the vane gears 164 are arranged in the exterior area 136 of the vane housing assembly 106, and each of the plurality of vanes 160 is arranged within the fluid flow passageway P. Accordingly, in the illustrated embodiment, each of the vane gears 164 are accessible to an operator for inspection and/or repair without needing to disconnect or disassemble the first housing portion 102 and the second housing portion 104. In embodiments in which the vane gear 164 is removably connected to the stem 162, the vane gear 164 can be readily replaced with another vane gear. For example, an operator may replace a worn or damaged vane gear 164 by disconnecting the vane gear 164 from the stem 162 and connecting a new or repaired vane gear 164 to the stem 162. As mentioned above, the vane gear 164 and the stem 162 may be formed of a single piece. Furthermore, disconnecting the first housing portion 102 from the second housing portion 104 provides access to all of the plurality of guide vanes 112, simultaneously.

FIG. 14 is a perspective view of the ring gear 108. The ring gear 108 includes a first surface 206, a ring second surface 208, and an annular wall 210 extending therebetween. The annular wall 210 includes a ring inner surface 212 defining the boundary of a ring opening 214. The inner surface 212 is sized and shaped such that at least a portion of the first housing portion 102 can be received within the ring opening 214. In the illustrated embodiment, the ring gear 108 is rotatably connected to and rotatable relative to the first housing portion 102. In other embodiments, the ring gear 108 is rotatably connected to the second housing portion 104.

The first housing portion 102 includes a lip 216 extending radially outward from the first outer surface 130 (FIG. 13). The lip 216 engages a surface 217 of the press fit bearing 110 when the bearing 110 is disposed around the first outer surface 130. The lip 216 engages the bearing 110 to prevent or inhibit axial translation of the bearing 110 relative to the first housing portion 102. Alternatively, when the ring gear 108 is rotatably connected to the first housing portion 102, the lip 216 may be in contact with the first surface 206 of the ring gear 108 to prevent or inhibit translation axially of the ring gear 108 relative to the first housing portion 102.

FIG. 15 is an end view of the ring gear 108 and FIG. 16 is a side view of the ring gear 108. The ring second surface 208 of the ring gear 108 includes a plurality of gear teeth 218. The gear teeth 218 of the ring gear 108 are sized and shaped to mate with the gear teeth 186 of each of the vane gears 164 of the guide vanes 112. Rotation of the ring gear 108, about the housing axis A_{106} , is transmitted to the vane gears 164, causing rotation of the guide vanes 112 about the vane axis A_{162} , within the guide vane openings 200 of the vane housing assembly 106. The ring gear 108 further includes a feature 222 disposed on the inner surface 212. The feature 222 engages with the bearing 110, preventing the bearing 110 from translating axially relative to the ring gear

12

108. The feature 222 may span about the circumference of the inner surface 212. In some example embodiments, the feature 222 includes a lip.

The ring gear 108, and the vane gears 164, are arranged in the exterior area 136 of the vane housing assembly 106 enabling an operator to inspect and/or repair the ring gear 108 without disconnecting the first housing portion 102 and second housing portion 104. The accessibility of the ring gear 108 and the vane gears 164 is beneficial in that it reduces the time required to inspect and/or repair the ring gear 108 and the vane gears 164. In addition, the accessibility of the vane gears 164 and the ring gear 108, without disassembling the first housing portion 102 and the second housing portion 104, helps prevent bearing surfaces between the first channel surface 196, the second channel surface 148, and the stem 162 from being exposed to debris and/or contaminants.

In the illustrated embodiment, at least one of the guide vanes 112 is a drive guide vane 220. The drive guide vane 220 is operably connected to motor 174 (e.g., by the driveshaft 175), and the motor drives rotation of the drive guide vane 220. Rotation of the drive guide vane 220 causes rotation of the ring gear 108, which transmits rotation to the rest of the plurality of guide vanes 112, referred to as follower guide vanes. Accordingly, all of the plurality of guide vanes 112 rotate in unison. In some embodiments, the motor is a stepper motor. In some embodiments, the motor is communicatively connected to a controller and the controller transmits one or more instructions to the motor causing the motor to rotate the drive guide vane 220 in order to arrange the plurality of guide vanes 112 in a selected orientation relative to the fluid flow F.

FIG. 17 is a perspective view of the bearing 110. The bearing 110 may be arranged between the first outer surface 130 of the first housing portion 102 and the ring inner surface 212 of the ring gear 108. The bearing 110 facilitates rotation of the ring gear 108 about the first housing portion 102. In some embodiments, the bearing 110 is connected to the ring gear 108, e.g., the bearing 110 is press fit into frictional engagement with the ring inner surface 212. Accordingly, the ring gear 108 and the bearing 110 rotate relative to the first housing portion 102. Alternatively, the bearing 110 may be press fit onto first housing portion 102 such that the bearing 110 and the first housing portion 102 are frictionally engaged and the ring gear 108 rotates relative to the bearing 110 and the first housing portion 102.

The bearing 110 may be a non-lubricating bearing or a self-lubricating bearing. As such, the bearing 110 does not require application of lubricants. In some embodiments, for example, the bearing 110 is constructed of bronze and/or bronze composite. In some embodiments, the bearing 110 is bronze coated. The bearing 110 may be impregnated with lubricants or the bearing 110 may include one or more graphite plugs. In alternative embodiments, the bearing 110 may include any suitable type of bearing 110 that enables the inlet guide 100 to function as described herein.

In alternative embodiments, the bearing 110 may be omitted and the ring gear 108 may rotate about the first housing portion 102, without the use of a bearing.

FIG. 18 is a rear view of the inlet guide 100, having the first housing portion 102, ring gear 108, and bearing 110 removed therefrom to illustrate the arrangement of the plurality of guide vanes 112 arranged within the plurality of second channels 150. As described above, simultaneous rotation of the plurality of the guide vanes 112 changes the orientations of the vanes 160 relative to fluid flow F entering the inlet 134. For example, the guide vanes 112 may be

13

rotated, in unison, to arrange the inlet guide **100** to any suitable position based on the operational needs of the compressor. For example, the guide vanes **112** may be rotated, in unison, to arrange the inlet guide **100** to a fully open or neutral position. In the fully open position, the guide vanes **112** are arranged such that the first vane side **166** and the second vane side **168** of the vanes **160** are arranged generally parallel to the direction of the fluid flow **F** such that the vanes **160** do not substantially impede the fluid flow **F** through the fluid flow passageway **P**. In the fully open position, the vanes **160** align the fluid flow **F** creating a more laminar fluid flow **F** profile thereby increasing efficiency of the compressor. The plurality of guide vanes **112** may be rotated to arrange the vanes **160** in any suitable orientation relative to the fluid flow **F** and the fluid flow passageway **P**. For example, the motor may rotate the drive guide vane **220** in either a clockwise or a counterclockwise direction to adjust the orientation of the plurality of vanes **160**. The position of the vanes **160** may be selected to increase the operating range of the compressor, including both surge and choke.

In some embodiments, sensors (not shown) may be attached to one or more of the guide vanes **112** to measure the rotational position of the guide vanes **112**. The sensor may be communicatively coupled to the controller. The controller may use feedback received from the sensor to determine instructions for the motor. In some embodiments, one or more sensors may be used to measure the rotational speed of the guide vanes **112**. For example, the guide vanes **112** may be rotated at a rotational speed of 1 rotation per min (rpm) or less.

FIGS. **19** and **20** are perspective and sectional views, respectively, of a compressor **300** suitable for use with the inlet guide **100** described herein. The compressor **300** is illustrated in the form of a two-stage centrifugal compressor. The compressor **300** generally includes an outer compressor housing **302** forming at least one sealed cavity within which each stage of refrigerant compression is accomplished. The compressor **300** includes a first refrigerant inlet **310** to introduce refrigerant vapor into the first compression stage, a first refrigerant exit **314**, a refrigerant transfer conduit **312** to transfer compressed refrigerant from the first compression stage to the second compression stage, a second refrigerant inlet **318** to introduce refrigerant vapor into the second compression stage, and a second refrigerant exit (not shown in FIG. **20**). The inlet guide **100** may be arranged in proximity to the first refrigerant inlet **310**. For example, the outlet **132** to the fluid flow passageway **P** may be arranged in alignment with the first refrigerant inlet **310**. Refrigerant transfer conduit **312** is operatively connected at opposite ends to the first refrigerant exit **314** and the second refrigerant inlet **318**, respectively. The second refrigerant exit delivers compressed refrigerant from the second compression stage to a cooling system in which compressor **300** is incorporated. The refrigerant transfer conduit **312** may further include a refrigerant bleed (not shown in FIG. **20**) to add or remove refrigerant as needed at the compressor **300**.

Referring to FIG. **20**, the outer compressor housing **302** encloses a first compression stage **324** and a second compression stage **326** at opposite ends of the compressor **300**. The first compression stage **324** includes a first stage impeller **306** configured to impart kinetic energy to incoming refrigerant gas entering via the first refrigerant inlet **310**. The kinetic energy imparted to the refrigerant by the first stage impeller **306** is converted to increased refrigerant pressure (i.e., compression) as the refrigerant velocity is slowed upon transfer to a diffuser formed between a first stage inlet ring

14

301 and a portion of the outer compressor housing **302**. Similarly, the second compression stage **326** includes a second stage impeller **316** configured to add kinetic energy to refrigerant transferred from the first compression stage **324** entering via the second refrigerant inlet **318**. The kinetic energy imparted to the refrigerant by the second stage impeller **316** is converted to increased refrigerant pressure (i.e., compression) as the refrigerant velocity is slowed upon transfer to a diffuser formed between a second stage inlet ring **303** and a second portion of outer compressor housing **302**. Compressed refrigerant exits the second compression stage **326** via the second refrigerant exit (not shown in FIG. **20**).

The first stage impeller **306** and second stage impeller **316** are connected at opposite ends of a driveshaft **304** that rotates about a driveshaft axis A_{304} . The driveshaft extends from a driveshaft first end **330** to a driveshaft second end **332** and is axisymmetric about the driveshaft axis A_{304} . Additionally, the driveshaft axis A_{304} extends through a center of gravity of the driveshaft **304**. The driveshaft **304** is operatively connected to a motor **308** positioned between the first stage impeller **306** and second stage impeller **316**, such that the motor **308** rotates the driveshaft **304** about the driveshaft axis A_{304} . The first stage impeller **306** and the second stage impeller **316** are both connected to the driveshaft **304** such that the first stage impeller **306** and second stage impeller **316** are rotated at a rotation speed selected to compress the refrigerant to a pre-selected pressure exiting the second refrigerant exit. Any suitable motor may be incorporated into the compressor **300** including, but not limited to, an electrical motor.

Embodiments of the inlet guide vane apparatus described herein require simple assembly of a few number of components. Embodiments of the inlet guide vane apparatus, described above, include a first and second housing portion, a ring gear and bearing, and a plurality of guide vanes. In the illustrated embodiments, the vane gear of the inlet guide vane and the ring gear are both arranged in an area exterior to the vane housing. Accordingly, these components are accessible to an operator to perform and inspection and/or a repair operation. Specifically, the operator does not need to disassemble the inlet guide vane apparatus to inspect and/or replace the gears of the guide vanes. In one illustrated embodiment, the vane gear is selectively connected to the stem of the guide vane. Accordingly, the vane gear may be removed and replaced, without requiring the operator to disassemble the housing minimizing exposure of bearing surfaces to contaminants. Embodiments of the bearing may be a non-lubricating bearing or is a self-lubricating bearing. As such, the bearing does not require application of lubricants for operation. Embodiments described herein include a relatively small number of components, e.g., a two part "split-housing" that allows the operator to quickly (e.g., within 10 minutes) disassemble and/or reassemble the inlet guide vane apparatus. Furthermore, an operator may only need to use a single tool to disassemble and/or reassemble the inlet guide vane.

As used herein, the terms "about," "substantially," "essentially" and "approximately" when used in conjunction with ranges of dimensions, concentrations, temperatures or other physical or chemical properties or characteristics is meant to cover variations that may exist in the upper and/or lower limits of the ranges of the properties or characteristics, including, for example, variations resulting from rounding, measurement methodology or other statistical variation.

When introducing elements of the present disclosure or the embodiment(s) thereof, the articles "a," "an," "the" and

15

“said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” “containing” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. The use of terms indicating a particular orientation (e.g., “top,” “bottom,” “side,” etc.) is for convenience of description and does not require any particular orientation of the item described.

Having thus described several illustrative embodiments, it is to be appreciated that various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to form a part of this disclosure, and are intended to be within the spirit and scope of this disclosure. While some examples presented herein involve specific combinations of functions or structural elements, it should be understood that those functions and elements may be combined in other ways according to the present disclosure to accomplish the same or different objectives. In particular, acts, elements, and features discussed in connection with one embodiment are not intended to be excluded from similar or other roles in other embodiments. Additionally, elements and components described herein may be further divided into additional components or joined together to form fewer components for performing the same functions. Accordingly, the foregoing description and attached drawings are by way of example only and are not intended to be limiting.

What is claimed is:

1. An inlet guide vane apparatus, the apparatus comprising:
 - a housing assembly defining a fluid flow passageway, the housing assembly comprising:
 - a first housing portion; and
 - a second housing portion connected to and positioned axially upstream of the first housing portion, wherein the first and second housing portions cooperatively define a plurality of guide vane openings extending radially through the housing assembly and into the fluid flow passageway, wherein at least one of the first housing portion and the second housing portion defines at least one slot, the slot extending from a first stop surface to a second stop surface;
 - a ring gear rotatable relative to the housing assembly; and
 - a plurality of guide vanes connected to the housing assembly, each guide vane comprising:
 - a stem extending from a first end to a second end, wherein at least a portion of the stem is disposed within one of the guide vane openings, wherein at least one of the guide vanes includes a stop extending from the stem and being sized and shaped to be received in the slot;
 - a vane gear disposed at the first end of the stem and operably connected to the ring gear; and
 - a vane disposed at the second end of the stem and within the fluid flow passageway of the housing assembly, wherein each of the guide vanes is rotatable relative to the housing assembly such that an orientation of the vane within the fluid flow passageway is selectively adjustable.
2. The inlet guide vane apparatus of claim 1, wherein the ring gear and the vane gear are accessible from an exterior of the housing assembly.
3. The inlet guide vane apparatus of claim 1, wherein the first housing portion defines a plurality of first channels and the second housing portion defines a plurality of second channels, wherein each of the first channels is aligned with

16

one of the second channels such that the first and second channels cooperatively define the plurality of guide vane openings.

4. The inlet guide vane apparatus of claim 1 further comprising a bearing, wherein the ring gear is rotatably connected to the housing assembly by the bearing.

5. The inlet guide vane apparatus of claim 4, wherein the bearing is a self-lubricating bearing.

6. The inlet guide vane apparatus of claim 1, wherein the plurality of guide vanes includes a drive guide vane and a plurality of follower guide vanes.

7. The inlet guide vane apparatus of claim 6, wherein the drive guide vane is operably connected to a motor operable to rotate the drive guide vane, wherein rotation of the drive guide vane drives rotation of the ring gear and rotation of the ring gear imparts rotation to the plurality of follower guide vanes.

8. The inlet guide vane apparatus of claim 7, wherein the motor is a stepper motor, and wherein contact between the stop and at least one of the first and second stop surfaces stops rotation of the plurality of guide vanes and stops the motor.

9. The inlet guide vane apparatus of claim 1, wherein each guide vane includes the stop extending radially outward from the stem.

10. The inlet guide vane apparatus of claim 1, wherein the vane gear of each guide vane is removably connected to the guide vane.

11. A compressor comprising:

- a compressor housing including an inlet;
- a driveshaft rotatably supported within the compressor housing;
- an impeller connected to the driveshaft and operable to impart kinetic energy to incoming refrigerant gas upon rotation of the driveshaft;
- an inlet guide vane apparatus connected to the compressor inlet, the inlet guide vane apparatus comprising:
 - a housing assembly defining a fluid flow passageway, the housing assembly comprising:
 - a first housing portion; and
 - a second housing portion connected to and positioned axially upstream of the first housing portion, wherein the first and second housing portions cooperatively define a plurality of guide vane openings extending radially through the housing assembly and into the fluid flow passageway, wherein at least one of the first housing portion and the second housing portion defines at least one slot, the slot extending from a first stop surface to a second stop surface;
 - a ring gear rotatable relative to the housing assembly; and
 - a plurality of guide vanes connected to the housing assembly, each guide vane comprising:
 - a stem extending from a first end to a second end, wherein at least a portion of the stem is disposed within one of the guide vane openings, wherein at least one of the guide vanes includes a stop extending from the stem and being sized and shaped to be received in the slot;
 - a vane gear disposed at the first end of the stem and operably connected to the ring gear; and
 - a vane disposed at the second end of the stem and within the fluid flow passageway of the housing assembly, wherein each of the guide vanes is rotatable relative to the housing assembly such that an orientation of the vane within the fluid flow passageway is selectively adjustable.

17

12. The compressor of claim 11, wherein the ring gear and the vane gear are accessible from an exterior of the housing assembly.

13. The compressor of claim 11, wherein the first housing portion defines a plurality of first channels and the second housing portion defines a plurality of second channels, wherein each of the first channels is aligned with one of the second channels such that the first and second channels cooperatively form the plurality of guide vane openings.

14. The compressor of claim 11, further comprising a bearing, wherein the ring gear is rotatably connected to the housing assembly by the bearing.

15. The compressor of claim 14, wherein the bearing is a self-lubricating bearing.

16. The compressor of claim 11, wherein the plurality of guide vanes includes a drive guide vane and a plurality of follower guide vanes.

17. The compressor of claim 16, wherein the drive guide vane is operably connected to a motor operable to rotate the drive guide vane, wherein rotation of the drive guide vane drives rotation of the ring gear and rotation of the ring gear imparts rotation to the plurality of follower guide vanes.

18. The compressor of claim 17, wherein the motor is a stepper motor, and wherein contact between the stop and at least one of the first and second stop surfaces stops rotation of the plurality of guide vanes and stops the motor.

19. A method of assembling an inlet guide vane apparatus including a housing assembly and a plurality of guide vanes, wherein the housing assembly includes a first housing portion and a second housing portion, and wherein each

18

guide vane includes a stem extending from a first end to a second end, a first vane gear disposed at the first end of the stem, and a vane disposed at the second end of the stem, and wherein at least one of the guide vanes includes a stop extending from the stem, the method comprising:

rotatably connecting a ring gear to the first housing portion;

connecting the first housing portion to the second housing portion such that the second housing portion is positioned axially upstream of the first housing portion, and such that the first and second housing portions cooperatively define a plurality of guide vane openings extending radially through the housing assembly and wherein at least one of the first housing portion and the second housing portion defines at least one slot; and positioning the plurality of guide vanes relative to the second housing portion such that, when the first housing portion is connected to the second housing portion, at least a portion of the stem of each guide vane is disposed within one of the guide vane openings and the stop is disposed within the slot.

20. The method of claim 19, wherein the first vane gear of each guide vane is removably connected to the stem, wherein the method further comprises:

removing the first vane gear from the stem of one of the guide vanes; and

connecting a second vane gear to the stem of the one guide vane, without disconnecting the first housing portion from the second housing portion.

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